

VARPA: A Variable Power Amplifier



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Abstract

VARPA: A Variable Power Amplifier

Having the capability to vary the RF output power of a wireless transmitter has become a key requirement for users and, by extension, RF system designers. This requirement is primarily driven by external factors such as energy efficiency or RF output power constraints.

Furthermore, wireless systems operating in the VHF band are of general interest for long-range, license-free telemetry, since the VHF band offers a unique set of advantages over more traditional, high-frequency bands such as UHF. These advantages include: design using discrete/lumped components and verification using low-cost RF test equipment. VHF electromagnetic propagation characteristics are also more favorable for large, open terrain, which is commonly found in rural South Africa.

This dissertation will cover the research, design and verification of a 0 to 1 watt (30 dBm) software adjustable RF power amplifier (PA) operating in the VHF band. The RF power amplifier design emphasizes efficiency and linearity while balancing the trade-off between power step size and system cost. To this end, a closed-loop control system is designed and implemented that uses a directional coupler and an RMS logarithmic power detector, in a feedback configuration, to allow the RF output power to be adjusted in step sizes of approximately 0.5 dB. Two power control strategies were investigated, namely: variable inline attenuation, which involves placing a variable digital attenuator at the input of the PA, as well as drain bias control which involves varying the supply voltage to the RF power amplifier itself.

The resulting system consists of four distinct modules. These include the power supply module which provides the required energy to the rest of the system and regulates the PA supply voltage, the transceiver module which generates the required PA input tone, the power amplifier module which regulates the RF output power of the system, as well as an auxiliary controller module which acts as the global system coordinator. Through a high-level and detailed design and simulation stage as well as the physical construction of mixed signal PCBs, a functional system prototype was successfully developed.

The prototype of the resulting system was able to achieve a PAE efficiency of 54% in class-AB mode and a large signal gain of 10.89 dB at a maximum RF output power of +30 dBm. Most importantly, the system prototype was able to set the RF output power from 0 to 30 dBm in 0.5 dB steps, with an error less than 20%, via a serial interface. The total component cost of the drain bias power control circuitry was also optimized, with a total BOM cost of less than \$3 for the external power control circuit. Overall, the system was able to achieve all but one of the research and design objectives, which was a 60% PAE efficiency target due to non-ideal Q for the filtering and matching networks as well as operation below the P1dB compression point of the RF LDMOS MOSFET.

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1 Introduction

1.1 Background

Radio frequency (RF) power amplifiers (PAs) are used in a wide variety of telecommunication systems, including but not limited to: mobile satellite terminals (Figure 1.1), cell towers and wireless routers. The main purpose of an RF power amplifier is to increase the AC power of an RF signal while retaining its original spectral content. More generally, we are interested in amplifying the real component of the signal power.



Figure 1.1: Power amplifiers are used in a wide range of communication equipment. Shown here is the commercial Mantis satellite terminal [5].

PA design techniques have evolved significantly over the past 50 years. If anything, the emergence of non-linear simulation techniques such as harmonic balance (HB) and accurate non-linear simulation models of RF power transistors has provided engineers with extremely powerful tools for the design and analysis of RF power amplifiers (PAs).

The aim of this dissertation is to develop and characterize circuitry that can be added to a traditional class-AB RF power amplifier design with as little modification to the PA as possible. The added circuitry should then allow the desired RF output power to be set via a software interface with sufficient resolution.

1.2 Motivation

Power amplifiers are a critical part of a radio-frequency communications chain. Without them, a radio transmitter may lack sufficient RF power to transmit information to a distant receiver, essentially rendering the entire communications chain ineffective.

Single-ended class-AB PAs typically operate at a fixed output power, near the P1dB compression point, where power added efficiency (PAE) and output power are maximized [6] (Figure 4.18). However, there are many applications that require variable RF output power. For example, there may be region-specific regulatory requirements or limits on RF transmit power (i.e., FCC, ICASA, EU CEPT), requiring different levels of transmit power for a specific RF device class (such as a short range device). An RF designer might want to retain high-output power capability while meeting these regulatory requirements in each region using a firmware setting, for example.

Another important consideration for battery-powered transmitters is power consumption. Higher power consumption means less on-air transmit time. Therefore, dynamic control of the RF output power of a PA is one of two important considerations (power and time) in improving energy efficiency and system uptime, since the PA stage is often the main power consumer in a wireless system.

Therefore setting the desired output power or transmission time in firmware/software is often the most useful approach in practice, since modern RF systems often integrate some type of programmable controller. Examples of software-driven RF power amplifiers include software-defined radio (SDR), mobile handsets [7], modern radar systems and satellite communication terminals.

1.3 Objectives

The main research goal of this dissertation is to design and build both a standalone class-AB VHF power amplifier and associated control circuitry that allow the RF output power of the PA to be set digitally. This involves investigating two different power control techniques. The two power control strategies that will be investigated are variable drain voltage bias and attenuation-based power control.

Additionally, the broad design objective of this dissertation will be to develop a closed-loop RF output power control system, utilizing feedback from an RF RMS power detector, for an LDMOS (laterally diffused metal-oxide semiconductor) RF power MOSFET and then to determine the efficiency, gain, and step resolution of the system prototype.

Due to the nature of VHF electromagnetic propagation, the system, when integrated with a radio transceiver, is ideal for use in large open spaces, such as those found in rural areas. The benefits include being able to maximize the power efficiency of the system, as the cost of utility power is generally a limiting factor for rural communities. The system could also potentially be used in community base stations for metering applications or to transmit sensor data from weather beacons, for example.

The design requirements for the power amplifier module are derived from common FCC and EU regulatory performance standards such as FCC 47 CFR 97.307 [8], ETSI EN 300-220 [9] and ERC 70-03 [10] for RF SRDs (short range devices) and serve as typical values for the design.

The PA design requirements are:

- 169.4 MHz (VHF) operating frequency.
- Single-ended, class-AB mode of operation (low cost).
- 60 % PAE or greater (a reasonable efficiency target for class-AB).
- Stable output ($K > 1$) for design and harmonic frequencies.
- Maximum RF output power of at least 1W (+30 dBm).
- Minimum large signal gain (G_{LS}) of at least +10 dB.
- 50 dBc of harmonic attenuation at +30 dBm of output power by PA output filtering network.
- Support for FSK (non-AM) modulation.

In terms of RF power control, the design requirements are as follows:

- 0 to 30 dBm RMS output power range.
- 0.5 dB power step size resolution, within 20 % error.
- < 1.5 dB absolute amplitude accuracy of RF output power.
- Power step ramp settling time < 1s.

A design methodology is required to synthesize the design from the requirements identified above. The approach taken in the dissertation will be to divide the design work into a high-level and a detailed design and simulation stage.

1.4 Methodology

A six-stage methodology was developed as a formal guide for the entire dissertation and is described below. The methodology emphasizes research and simulation to ensure that the design of the system prototype follows best practices and is well optimized.

1.4.1 Stage 1 : Literature Review

The initial research stage aims to develop a modern, theoretical framework for the high-level and detailed design sections of the dissertation. It consists of a review of RF power amplifier design and power control literature which is required for the design of the power amplifier module.

1.4.2 Stage 2 : Review and Planning

Once a broad theoretical framework is developed, preparation for design work then follows in Stage 2. As with any engineering design process, it is expected that multiple design iterations will be required before the final hardware and software prototypes are developed. This fact should be taken into account in the project plan.

1.4.3 Stage 3 : High-Level Design

The high-level design stage aims to develop the necessary subsystem block diagrams for each system module. This is an important stage in the design process and allows a designer to evaluate the required input and output signal relationships between different modules at a high level before starting detailed design work.

1.4.4 Stage 4 : Detailed Design and Simulation

Once the necessary block diagrams have been developed, each module can be synthesized by applying the relevant design procedures. For each hardware module, a circuit schematic and PCB layout is synthesized, while for software modules the necessary C source code is written.

The design of each system module should first be validated through simulation before the physical design and construction of the PCB begins. Once the theoretical design is validated, components and PCBs can be ordered and a physical system prototype constructed.

1.4.5 Stage 5 : Measurements and Analysis

After the construction of a system prototype, measurements are required to determine if the final design meets the design requirements set out in the dissertation. This stage involves the use of laboratory measurements/procedures to determine important performance parameters. In addition, the results of laboratory experiments are then collected and analyzed to determine if the research and design objectives were fully met.

1.4.6 Stage 6 : Conclusions

This section of the dissertation focuses on design limitations, future improvements, and future work, as well as an overall analysis of the dissertation and its outcomes.

2 Literature Review

This chapter will review the available academic literature relating to RF power amplifier design with the aim of developing a broad theoretical framework that will be used to guide the design process.

The literature review will begin with a review of past PA design techniques followed by more modern PA design methods. The different RF power transistor technologies will then be investigated, with the aim of choosing the correct technology for VHF. PA stability and efficiency measures will then be investigated, followed by a review of the RF PA output power control methods. Finally, the literature review will conclude with a brief review of RF modulation schemes as well as ICASA regulations for non-specific SRD's operating in the VHF band.

2.1 Historical RF PA Design Techniques

Using a well-characterized and proven PA design technique is a critical requirement for a robust PA design. Historically, two main methods of PA design were in widespread use. In the first method, a passive variable impedance network or mechanical 'tuner' (typically a coaxial stub tuner) is placed at the input and output ports of the RF power transistor [11] or DUT (Figure 2.1). The impedance of these tuning networks is then manually varied until a specific operating condition (i.e output power or efficiency) is obtained through direct measurement of the DUT in a process known as 'load-pull'.

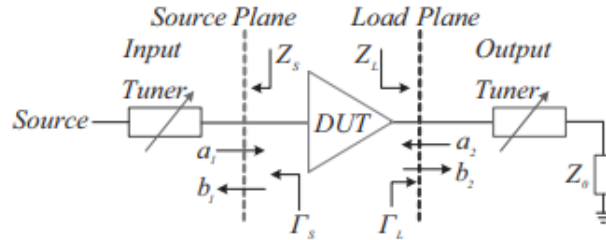


Figure 2.1: Traditional power amplifier load-pull setup with physical impedance tuners [1].

As nonlinear devices, the optimal loading conditions of RF power transistors differ significantly from the conjugate match obtained by linear analysis of the S parameters [12]. Despite this, it has been observed that small-signal S-parameters measured at the input port of an RF power transistor, operating at a specific output power level, can provide a good starting point for the design of the PA input matching network [13] when a non-linear model is not available to a designer.

The Cripps method (CM) [14] is a historical design method that was devised [11] in order to provide designers with a more accurate analytical PA design technique over older, less accurate methods that were used at the time. In the Cripps method, the output of an RF power transistor is modeled as a current source in parallel with the output capacitance of the transistor (C_{out}) [15] with a load impedance Z_L connected in parallel.

The Cripps method determines the optimal load impedance for maximum output power based on the DC I-V characteristics of the transistors and the small-signal output impedance at a given DC

operating point [16]. The Cripp's method has the advantage of being straightforward to use and generally works well for single-tone applications. However, the precision of these early analytical techniques was highly dependent on having a good empirical model of the corresponding RF power transistor [12]. It should be noted that these older methods of PA design do not provide the same benefits of more modern, simulation-based techniques.

2.2 Modern RF PA Design Methods

With more research on PA design being conducted over time, new techniques have emerged that take advantage of the exponential increase in computing power over the last decade (Moore's law). However, a basic background to the PA design process is a prerequisite to understand these modern PA design techniques.

Under large signal conditions, the performance (i.e. gain, efficiency) of a power amplifier depends not only on the frequency and bias set point of the power transistor, but also on the input/output power and PA loading conditions [17]. The load-pull technique has emerged as the most widely used design technique to optimize PA performance in the non-linear domain [18]. Recent advances in commercial automatic load-pull sets and RF simulation tools have significantly simplified the traditional RF power amplifier design process.

In particular, the emergence of the harmonic balance (HB) method and its mathematical variants has allowed for the development of large-signal simulated load-pull techniques [19] [15], which have largely replaced traditional mechanical or automatic load-pull test equipment when an accurate RF power transistor software model is available to a designer. In instances where a particular RF power transistor software model is not available, a physical load-pull set is often required for a PA design.

Mechanical load-pull systems can be classified as active or passive or as a combination of both. In modern passive systems, the desired impedance is synthesized by controlling the length and position of impedance tuning elements/stubs using a mechanical mechanism [20]. In active load-pull systems, the desired impedance is synthesized electronically by controlling the phase and amplitude of an injected RF signal with a phase-synchronized signal source. Active load pull systems have the advantage of being able to synthesize a very small and accurate impedance [20] but are typically much more expensive than passive systems. However, in general, modern commercial load-pull sets are often prohibitively expensive for the average student or hobbyist.

A low cost, mechanical impedance tuner utilizing a 3 dB quadrature hybrid directional coupler and two shorted microstrip transmission line segments has been proposed in the literature [21]. The design is simple to implement, but the physical length of the transmission lines and coupler is dependent on the design frequency of the impedance tuner which makes the tuner more suitable for high band VHF and UHF load-pull.

2.2.1 Harmonic Balance

Harmonic balance (HB) is a novel iterative frequency domain technique used to calculate the steady-state response of a nonlinear circuit [22] [23]. The HB method makes use of the fact that periodic waveforms in many circuits are best described in terms of their Fourier coefficients [24].

Given a discrete voltage waveform x_n , which contains M uniformly spaced samples, then:

$$x_n = \sum_{k=0}^{M-1} X_k e^{2\pi j \frac{kn}{M}} \quad (1)$$

$$n = 0, \dots, M - 1$$

Here n represents the sample index and k is the frequency index. The goal of harmonic balance then is to find the Fourier coefficients X_k of all the conduction waveforms (voltage and current). These Fourier coefficients are related by KCL (Kirchoff's current law) in the frequency domain [24]. The target circuit can then be described in terms of a system of differential equations, derived from nodal analysis [25]:

$$f(x, t) + \frac{d}{dt}q(x, t) = 0 \quad (2)$$

Where q is a vector of all charges and fluxes entering/leaving a node and f is a vector of the conductive contributions to the circuit. Each circuit node contributes an equation arising from KCL. $\frac{d}{dt}q(x, t)$ is then discretized to produce a system of nonlinear equations which are processed further to obtain the KCL equations in the frequency domain.

These are then solved to obtain X_k , at which point the nonlinear frequency-domain response can be obtained using Equation 1 or processed further using the inverse Fourier transform to obtain the non-linear time-domain response of the circuit. Various commercial circuit simulators support the harmonic balance (HB) technique [26].

2.2.2 Multi-Tone Load-Pull

The second harmonic tone typically has a significant effect on the efficiency and linearity of an RF power amplifier. Traditional load-pull simulation techniques that rely on harmonic balance (HB) generally do not optimize the load for higher-order harmonics, which results in reduced efficiency [27] and linearity.

Typically a major problem with an RF power amplifier is out-of-band distortion (predominantly from the second harmonic) which gets mixed in to the fundamental band and is referred to as a 'memory effect' [28]. The optimization of harmonic impedances through harmonic load-pull is therefore crucial for reducing these memory effects and distortion. The standard HB technique does not provide information on how distortion is built up, only providing information on total distortion. A novel harmonic load-pull technique based on Volterra analysis has been proposed that aims to resolve these shortcomings [28].

Volterra analysis (VA) relies on a polynomial model of the transistor that can be easily fitted. VA has the advantage of having a faster analysis speed over harmonic balance (HB). It is worth noting that the load-pull using the HB method works similarly to traditional load-pull where the load reflection coefficient is systematically varied during simulation for a nonlinear input stimulus.

Another method of harmonic load-pull has been suggested, which is based on a numerical optimization algorithm that combines the benefits of Newton's method (NM) and the steepest decent method (SDM) and only requires first order derivative data [27]. Similarly to the Volterra

method, this method has the benefit of improved simulation speed over the harmonic balance method.

2.3 Lumped vs Distributed Element Networks

Since the system is operating in the VHF band, we can use discrete or lumped element components for the physical construction of the RF networks that are designed and simulated. This is typically less common for microwave frequencies as the parasitic effects of component leads and materials at such high frequencies cannot be ignored and, therefore, should be modeled during design.

The successful use of lumped components at RF frequencies below 500 MHz has been reported in the literature [29]. From around 500MHz to 1 GHz these lumped circuit elements or components become more difficult to design and model, and hence the use of distributed elements is preferred.

A typical practical limitation for lumped/discrete components is the self-resonant frequency (SRF) [30] and the Q-factor, which, in general, must be high enough for a particular application. Typically 'RF grade' capacitors and inductors are used to achieve a high Q and SRF at least 10 times higher than the desired RF operating frequency. For choking or bypassing applications, the requirements are not as strict during component selection. The advantage of lumped components is that they are often cheaper and easier to use in a circuit and to measure or characterize.

RF power transistors are typically supplied in through-hole or SMD (surface mound device) packages. Above HF it is more common to use SMD technology. The choice of RF transistor technology itself, however, is very important and is covered in the next section.

2.4 RF Power Transistor Technologies

LDMOS (laterally-diffused metal-oxide semiconductor) RF power transistors were initially introduced as a replacement for bipolar junction transistors (BJTs) in RF base station applications [31] and as a result are common in older designs. Prior to 2005, silicon (Si) LDMOS was the dominant technology for RF power applications below 3 GHz due to its low cost [32]. At the time the only competing technology was GaAS (Gallium Arsenide) but had the disadvantage of being more costly and being better suited towards lower power applications.

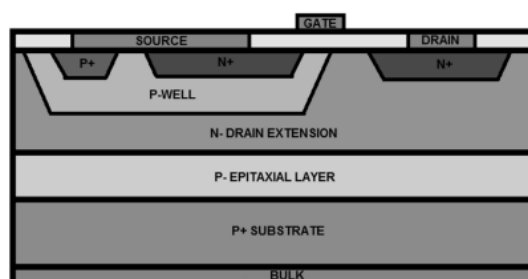


Figure 2.2: LDMOS MOSFET semiconductor topology [2].

Figure 2.2 shows the semiconductor layout of a typical LDMOS MOSFET transistor. These are 3 terminal devices (gate, drain, and source) in which a gate voltage controls the current flowing from the drain to the source. The introduction of GaN (Gallium Nitride)-based power

transistors mainly had the effect of providing higher power densities at higher frequencies (up to 5 times higher than LDMOS) to a designer [33] [32]. This allows an RF power transistor to operate at a higher power, frequency, and temperature, thereby simplifying the PA thermal design/management process. Power dissipation for these two technologies occurs primarily at the drain/source terminals, which are often connected to a heatsink.

LDMOS is still the leading technology for VHF power applications [31] due to high performance and low cost. Additionally, practical LDMOS PA circuits typically operate in class-A or class-AB mode [34] and a single-ended topology is usually preferred, again due to cost.

2.5 PA Stability Analysis

Stability analysis is a critical part of the RF PA design process [35]. Amplifiers can either be conditionally or unconditionally stable. A conditionally stable RF PA is prone to oscillation for certain load and source terminations [36] and requires the insertion of a stabilization network to dampen feedback [37].

A common historic, analytical stability measure used in PA design is the Rollet stability factor (K) [36] which is defined in Equation 3 and is based on S-parameter analysis (Equation 3):

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} \quad (3)$$

Here ($|\Delta|$) is defined as the scattering (S-parameter) matrix determinant (Equation 4):

$$|\Delta| = |S_{11}S_{22} - S_{21}S_{12}| < 1 \quad (4)$$

In order to ensure that an amplifier is unconditionally stable, we require that $K \geq 1$ [37].

Another analytical PA stability measure reported in the literature are the so called 'geometric' stability factors μ_{source} and μ_{load} [38] (defined in Equations 5 and 6):

$$\mu_{source} = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta(S_{11}^*)| + |S_{21}S_{12}|} \quad (5)$$

$$\mu_{load} = \frac{1 - |S_{22}|^2}{|S_{11} - \Delta(S_{22}^*)| + |S_{21}S_{12}|} \quad (6)$$

For unconditional stability the following must be satisfied (inequality 7 and 8):

$$\mu_{load} \geq 1 \quad (7)$$

$$\mu_{source} \geq 1 \quad (8)$$

Both Rollet (K) and geometric stability factors (μ_{source} , μ_{load}) rely on small-signal linear S-parameters to make predictions about the stability of the RF power amplifier. As large-signal nonlinear devices, S-parameters are not the most accurate mathematical tool to model PA

stability. In fact, the Rollet stability factor can have different values under different power and biasing conditions [36].

A novel method of stability analysis has been proposed that is based on the large signal S parameters [36]. Large-signal S-parameters are similar to traditional small-signal S-parameters but take into account the bias and power conditions of an RF power transistor [39]. PA stabilization networks can consist of a passive resistive element placed at the input or output of a PA or an RLC feedback network [37]. A RC stabilization network has been proposed in the literature with a method to determine component values using Y-parameters [40].

Another approach to power amplifier stability has been reported in the literature which replaces the traditional stability networks mentioned above. In the out-of-band feedback technique [35] a low frequency feedback network is embedded in the drain and gate bias regions of the RF power amplifier which produces a stable response up to the RF design frequency. This approach has the advantage of being free of lossy elements such as damping resistors and, as a result, maximizes the PA gain at the expense of increased design complexity and component count.

2.6 PA Efficiency Measures

The efficiency of the power amplifier is a critical performance parameter. A power amplifier that is inefficient tends to convert a large proportion of its total DC input power into heat, which is undesirable. Therefore, careful consideration must be taken when designing matching networks and selecting physical components, as these all have a direct effect on PA efficiency.

The total power consumed by an RF power amplifier may be determined by monitoring the DC current at the drain of an LDMOS power transistor. The total DC and AC power consumed is divided between the RF power transistor itself and the load (usually an antenna or 50 Ω dummy load). The two historic measures of RF power amplifier efficiency mentioned in the literature are *PAE* (Power Added Efficiency) and drain efficiency (η) [41]. *PAE* and drain efficiency (η) are defined by Equations 9 and 10:

$$PAE = \frac{P_{out} - P_{in}}{P_{DC}} \quad (9)$$

$$\eta = \frac{P_{out}}{P_{DC}} \quad (10)$$

Here, P_{in} represents the RF input power from a CW source, P_{out} represents RF power delivered to the load by the PA and P_{DC} represents the total DC power drawn from the DC supply. From the above equations, it can be seen that *PAE* is typically slightly lower than the drain efficiency (η). While these traditional measures of efficiency are sufficient for the majority of applications, they have a few drawbacks. In the case of a pulsed RF PA for example, the power dissipation is not constant over time, and therefore the average power/efficiency formulas mentioned above are not meaningful.

A novel method has been proposed for measuring the efficiency of PA's without constant power dissipation [42]. Using the proposed method, the DC input power is divided into several intervals in the time domain. The power source and the DUT are then fed with a pulse modulator, and the quiescent power is measured for each interval. The average efficiency is then measured with

constant input RF power. Finally, the efficiency parameters for each interval can then be extracted using the average efficiency and formulas provided in the research paper [42].

2.7 PA Compression Measurement Techniques

The P1dB compression point is a measure of the gain linearity of a power amplifier. It is the output power level at which an amplifier's gain drops by 1 dB from its nominal value, signaling the beginning of nonlinearity [43] [23] or gain compression. Most amplifiers start to compress (exhibit non-linear gain) approximately 5 to 10 dB below their P1dB point [44].

The P1dB point implies that RF power amplifiers typically have a linear and non-linear region of power gain, as shown in Figure 2.3. Traditionally, the P1dB compression point is measured using a spectrum or vector network analyzer (VNA). A novel technique has been proposed in the literature to measure the P1dB point of a PA integrated circuit internally and is based on the oscillation built-in self-test method (OBIST) [43].

Using this technique the RF power amplifier is set up to operate as an oscillator and produces a growing input amplitude through the placement of an inline amplifier, phase shifter and resistor into the feedback network. In addition, the OBIST method has the advantage of being low cost but requires a control circuit which results in increased circuit complexity. Other methods of measuring the P1dB compression point reported in the literature include using harmonic balance (HB) RF circuit simulators.

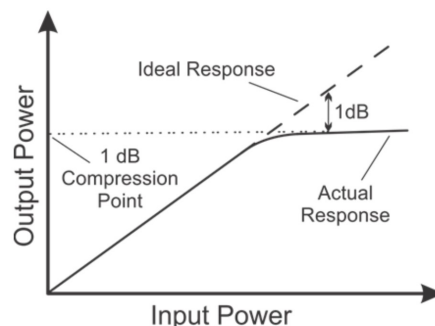


Figure 2.3: Definition of the P1dB compression point [3].

2.8 LDMOS Power Control Strategies

An RF power control strategy can either be open-loop or closed-loop and determines the mechanism that achieves RF output power control. Closed-loop power control strategies often make use of RF sensing elements such as a directional coupler and a RMS logarithmic power sensor [45] and are therefore more complex and costly to implement. The main drawback of the closed-loop approach is the insertion loss due to the RF directional coupling element. Additionally, the loop gain of the feedback network can vary significantly due to the non-linear characteristics of the PA [46], potentially introducing control stability issues. However, closed-loop power control generally has the advantage of improved accuracy compared to open-loop methods.

LDMOS power amplifiers have both a gate and a drain DC bias port. The traditional approach to RF power control reported in the literature for LDMOS power transistors has been to vary the

supply/drain bias [45] or the input drive using an inline attenuator at the PA input port.

Figure 2.4 depicts the proposed control circuit to vary the supply voltage of an RF power amplifier. It consists of an operational amplifier and power MOSFET that allows the supply voltage to be regulated linearly with a control voltage [46]. The maximum dynamic range of RF power control in dB using this strategy can be approximated using formula 11 [47] where DR represents the total dynamic range of RF power control (in dB) and $V_{DD,min}$ and $V_{DD,max}$ represent the minimum and maximum drain control voltages.

$$DR \approx 20 \log \frac{V_{DD,max}}{V_{DD,min}} \quad (11)$$

Reducing the voltage at the drain of an RF power MOSFET reduces the output swing, and therefore the power delivered to the load.

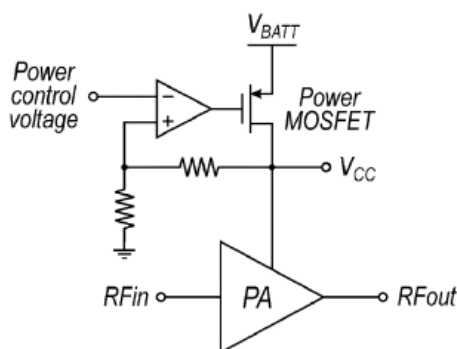


Figure 2.4: Drain bias control circuit proposed in the literature [4].

The RMS RF output power (P_{out}) of an RF PA in watts with a drain RFC (radio-frequency choke) can be estimated using the simple relation shown in Equation 12 [48] given the real load resistance (R_L), MOSFET saturation potential (V_{sat}) and drain bias (V_{DD}). This equation is generally valid when designing for the VHF band [49].

$$P_{out} = \frac{(V_{DD} - V_{sat})^2}{2R_L} \quad (12)$$

Another power control technique reported in the literature for LDMOS power transistors has been to vary the gate bias [50]. The disadvantage of this approach is that the operating (Q) point of the RF transistor will then change, affecting the stability and efficiency of the PA which results in an unstable closed-loop control response.

RF LDMOS power transistors are also susceptible to thermal runaway in which a large increase in temperature leads to a large increase in drain current. This feedback mechanism can lead to very high temperatures that can potentially damage the RF power transistor [50]. LDMOS transistors operate with a positive temperature characteristic near the quiescent point and thermal management requires more care than GaN transistors.

That is, for a fixed gate voltage the FET drain current (I_{DQ}) increases in proportion to the increase in temperature [34], while a decrease in temperature leads to a decrease in (I_{DQ}). Therefore,

some form of gate bias control and temperature compensation is required to set I_{DQ} and prevent thermal runaway regardless of the selected power control strategy. Temperature compensation can be achieved by monitoring the transistor's temperature or drain current [51].

It should be mentioned that the power control strategy that is selected has a direct impact on the total component cost of the control circuitry, and so the relevant tradeoffs must be weighed up appropriately before selecting a particular strategy.

2.9 LDMOS Gate Biasing Techniques

LDMOS RF power transistors require the application of a gate bias voltage (V_G) in order to set the quiescent or Q-point of the LDMOS MOSFET. The Q-point gate bias is usually determined by monitoring the drain current through the LDMOS MOSFET until a specific quiescent current is reached, at which point the gate voltage is recorded. It should be noted that the gate of a RF MOSFET transistor is high impedance which allows the gate voltage of the MOSFET to be set with a current limited DC voltage source, such as a DAC.

A gate bias control system has been proposed in the literature [52] which consists of an ARM processor, temperature sensor as well as a drain current monitoring circuit. The implementation makes use of ADC (analogue to digital converter) and a DAC (digital to analogue converter) to collect sensor data as well as set the gate bias which is required to reach the Q-point of the PA.

Achieving good linearity and efficiency at the same time in an RF PA design is typically difficult in practice [53]. Certain amplitude modulated schemes (such as QAM, OFDM) require a high degree of PA linearity and a number of different gate bias control schemes have been proposed in the literature to address these issues, increasing the efficiency of the PA at power levels below compression.

In the first technique, gate bias modulation (GBM), a higher efficiency can be achieved by continuously modulating the gate bias voltage as a function of the amplitude of the input RF signal [53]. Alternatively, instead of modulating the gate bias, the drain bias can be modulated instead in a technique known as envelope tracking (ET). In terms of PA linearization, the three main PA linearization techniques reported in the literature are: "predistortion", "feedback" and "feed-forward" [54] but these methods, while effective, are not applicable to this dissertation and will therefore not be covered in more detail.

2.10 Closed-Loop Control Algorithms

Various traditional control algorithms exist that can be used to control the output power of a PA or ensure a constant voltage bias. An approach to RF PA bias control in firmware has been proposed in the literature [34], which implements a form of PID (proportional-integral-derivative) control of the gate or drain bias voltage, with PI (proportional-integral) control being widely adopted in some RF power control system applications [55] [56] [57].

The PI controller has the advantage of being simple to implement [56] [58] but is not as robust at handling external disturbances [59]. A typical PID design procedure consists of first identifying the type of process model, designing the controller system, and then tuning the controller [58]. PID tuning is an integral part of PID controller design. Historical methods of PID tuning include: experimental trial and error, Ziegler-Nichols (ZN) and the Cohen-Coon (CC) method [60]. In

the ZN and CC methods, the tuning parameters are obtained for an operating point, where the control model can be evaluated linearly.

In some instances, these traditional methods of PID tuning do not yield good performance, and so more modern and sophisticated methods of PID tuning are required. In particular, the ZN and CC methods often do not typically yield the optimal PID tuning parameters [60] which leads to overshoots and oscillations in the response of the control system. A novel method of PID control tuning has been proposed in the literature which makes use of fuzzy logic [61] and artificial neural networks [62]. Research into AI optimization algorithms over the last decade has also yielded promising results, especially for PID control system tuning. PID controllers can be implemented discretely as a circuit or in software [63].

Other simpler methods of power control include linear voltage ramping [64] which can be just as effective as PID, provided that the ramp settling time is not too large for an end user.

2.11 Radio Modulation Schemes

For the purposes of transmitting data packets from one wireless node to another, various data modulation schemes exist. Some common radio modulation schemes include binary phase shift keying (BPSK), frequency shift keying (FSK) and amplitude shift keying (ASK).

2FSK is a digital modulation scheme in which an RF carrier is shifted between two discrete frequencies. It is defined by the peak frequency deviation (Δf) in Hz, which is equal to the difference between the high and low frequencies used in 2FSK [65] and the symbol rate (R_s) in symbols/sec which is the effective data transfer rate.

Carson's rule can then be used to calculate the total occupied bandwidth of the 2FSK signal [66] (Equation 13):

$$B_{2FSK} = 2(\Delta f + \frac{R_s}{2}) \quad (13)$$

Data rates of up to 1 Mbps using 2FSK can be achieved in practice [67] at the expense of greater bandwidth per FSK channel.

Turbo-FSK is a novel modulation technique proposed in literature that is designed for LPWA (low power wireless access) networks based on m-ary orthogonal frequency-shift keying (M-FSK) [68]. The technique has been shown to achieve a peak-to-average power ratio (PAPR) of around 0 dB with performance close to the Shannon limit for spectral efficiency and, as a result, achieves a higher data transfer rate than conventional FSK [69].

LoRa (short for "long range") is a proprietary and novel chirp spread-spectrum (CSS) modulation scheme designed for long-range, low data rate and low-power applications [70]. It is defined by a spreading factor (SF) [71] which determines the duration for which each symbol is transmitted. A larger SF typically results in an increased range. Another important parameter is the coding rate (CR) which is a measure of the degree to which error correction coding is applied to the transmitted data. Decreasing the coding rate generally results in an increase in range. In practice, LoRa can achieve data rates up to 250 Kbps.

2.12 ICASA Non-Specific SRD Regulations

Another aspiration of this dissertation is to produce an RF PA design that, through slight modification, can be commercialized and legally operated in the South African, US and EU regions. The commercial aim of the product would be to achieve long-range, high-power telemetry in rural terrain without the need for a radio frequency license.

It is quite common to find various low-cost RF transceivers operating in both regions illegally, and so a careful examination of the appropriate radio frequency legislation is required in order to determine the required transceiver operating parameters for the initial design.

The Independent Communications Authority of South Africa (ICASA) [72] is an independent body of the South African government responsible for national telecommunications apparatus regulation and enforcement. Furthermore, the 2015 Act on "Radio Frequency Spectrum Regulations" [73] is an important document regarding the use of unlicensed radio frequency transmitters. Specifically, amendment 737 of 2021 [74] to the radio frequency spectrum regulations defines which frequency bands and type of radio apparatus are exempt from radio frequency licensing in South Africa.

169.4-169.475M	Non-Specific SRD	500 mW e.r.p.
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Figure 2.5: ICASA regulations pertaining to short range devices operating in the license-free, 169.4 MHz VHF band.

Figure 2.5 is an extract from page 38 of the amendment.

It can be seen that the band from 169.4 to 169.475 MHz has been allocated to non-specific short range devices (SRDs) with a maximum ERP transmit power of 500 mW. Therefore, in a commercial version of the design, the maximum ERP transmit power must be limited to 500mW, which can potentially be implemented as a firmware feature.

Linear ERP transmit power is calculated using Equation 14 and is defined as the total power radiated by an actual antenna relative to a half-wave dipole.

$$ERP_{linear} \approx P_t \cdot \frac{G_t}{1.64} \quad (14)$$

Here, (P_t) represents the linear transmit power, while (G_t) represents the maximum linear gain of the antenna relative to an isotropic source. The Radio Frequency Spectrum Regulation Act of 2015 [73] defines the meaning of a non-specific SRD and is shown in Figure 2.6. The duty cycle limitation of 1 % is measured relative to a time window of 1 hour [75] and implies that the total transmit time in the 1-hour window is limited to 36 seconds. However, a duty cycle of 10 % is allowed for metering applications [76].

"Non specific Short Range Devices" means radio apparatus used for general telemetry, telecommand, alarms and data with a low duty cycle (<1.0%);

Figure 2.6: Formal definition of a short range device, according to ICASA.

ERC 70-03 [10] provides more information on the legal regulations of SRD devices in the EU region.

ERC 70-03 states that the 169.4 to 169.475 MHz band is available in most EU member states (with more details provided in the legal document) and has the same specifications and restrictions as defined by ICASA, with a duty cycle limitation of 10 %.

2.13 Conclusion

This section has reviewed past, present, and state-of-the-art literature concerning RF power amplifier design and output power control strategies. It has been determined that the load-pull technique will be required to determine the optimum input and output impedances of the selected RF power transistor and that LDMOS is the most suitable transistor technology for VHF, medium power (< 5W) applications.

Stability and efficiency analysis using the Rollet and geometric constants, as well as the drain and PAE (Power Added Efficiency) measures will be used to determine how well the power amplifier performs against core design requirements.

A suitable drain and gate biasing system using a DAC and microcontroller was also reviewed which can be used for biasing the gate of the RF MOSFET, as well as output power control strategies using variable drain/supply bias and inline input attenuation for varying the RMS RF output power of a PA.

Several power control algorithms were also investigated, ranging from PID control to simple linear ramping which will be used to handle sensor feedback and adjust either the PA drain voltage or input attenuation.

Next, various modulation schemes for the transmission of data packets were reviewed which will be used to verify that the prototype system transmits and receives FSK data without any issues and finally, a review of ICASA regulations applicable to non-specific short range devices (SRDs) was conducted in order to determine an upper limit on RF output power for the initial prototype as well as the required transmit duty cycle.

3 High-Level Design

With the literature review completed, the first iteration of the design can then proceed. The high-level design aims to identify, at a systems level, the required subsystems, their corresponding functionality as well as the input/output signal mappings.

A high-level block diagram of the proposed system is shown in Figure 3.1. It is seen to consist of four distinct blocks or 'modules'. The Tx/Rx (transceiver) module is responsible for generating the CW tone that is required to generate the necessary RF drive required by the PA module. The output of the transceiver module is then fed into a variable attenuator to the RF power amplifier (PA) module.

Within the PA module, the PA design is included along with the power detection circuitry. The combined auxiliary and power modules are then used to communicate between the digital sections of each system block, as well as to supply the variable drain bias to the PA, which is required to set the RF output power.

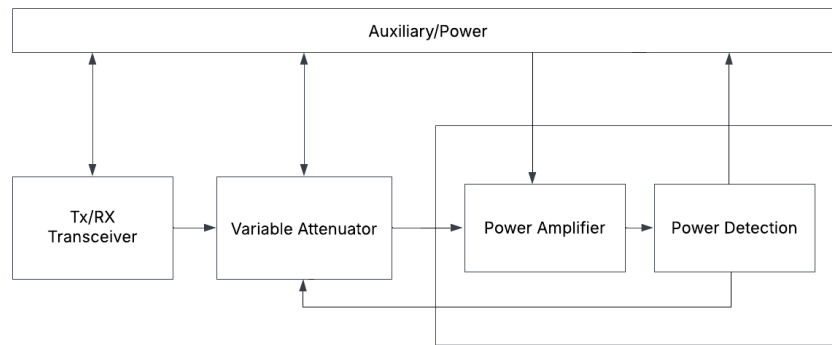


Figure 3.1: High-level overview of the required system modules, including the: transceiver, variable attenuator, power amplifier, power supply and auxiliary control modules.

3.1 Transceiver Module

3.1.1 Overview

The transceiver (TX) module is responsible for generating the CW (Continuous Wave) tone that is fed into the input of the PA module. The TX module is configured in firmware through a serialized data structure defined in a C header (.h) file that is sent to the radio during system startup. The transceiver also has the ability to transmit data packets.

Two common modulation schemes that are widely used for IoT applications are LoRa and 2FSK [77] [78]. LoRa is designed for low-power, long-range applications but suffers from low data rates. 2FSK is a simple modulation scheme which provides a moderate range at much higher data rates.

Commercial LoRa radio transceivers are currently manufactured by Semtech Corporation. A popular sub-GHz LoRa transceiver is the Semtech SX1261/2 [79]. However, the design of the PA and LNA matching networks for the SX1261/2 requires a load-pull instrument [80], making the design more complicated since a physical load-pull device was not available to the author.

There are many manufacturers of 2FSK RF transceivers. Silicon Labs offers a wide range of

sub-GHz 2FSK transceivers at an affordable price. The Si4468 [67] is a +20 dBm sub-GHz RF transceiver that supports a wide range of modulation modes such as: CW, FSK, GMSK and OOK. It features a low active power consumption of around 68mA in transmit (Tx) mode as well as a PA with a documented class-E matching procedure, which significantly simplifies the design of the transceiver's RF matching networks. Additionally, the Si4468 also includes a CW operating mode, which is required during the output power set-point ramping procedure.

3.1.2 Specifications

The simplest approach to complying with the 10 % duty cycle requirement identified in the literature review is simply to divide the 360 second available transmit time into equal intervals for the 1-hour window. Doing this requires us to estimate the total transmit time for each data packet. ERC 70-03 states that we are limited to a channel bandwidth of 50kHz which limits our maximum data rate. The total frequency deviation (Δf) and symbol rate (R_s) of a 2FSK modulated signal is related by the modulation index h (Equation 15) [81]:

$$h = 2 \cdot \frac{\Delta f}{R_s} \quad (15)$$

Equation 15 can then be rearranged to make Δf the subject of the formula and substituted into Equation 13 to find the maximum data rate given a minimum modulation index (h) of 0.5 (a standard minimum value for 2FSK) [78].

$$R_{s,max} = \frac{B_{2FSK}}{1.5} \approx 33 \text{ Kbps} \quad (16)$$

The frequency deviation Δf can then be calculated. Given a data rate of 33 Kbps, Δf is equal to 8.25 kHz. Since Δf can only have a single digit precision for the Si4468 transceiver, the final design Δf is set to 8 KHz, resulting in a symbol rate of 32 Kbps (4 KB/s) for a channel bandwidth of 48 KHz. Using Equation 13 (Carson's rule), the total occupied bandwidth is calculated as follows (which meets our requirements):

$$B_{2FSK} = 2(8000 + \frac{32000}{2}) = 48 \text{ KHz} \quad (17)$$

The next task is to calculate the total transmit time required per data frame. There is a trade-off between the total transmit time and the data frame size. A smaller total transmit time allows for faster updates at the expense of a reduced data frame size. Ideally we would like to transmit a data frame each second or faster.

A 30 ms transmit duration is required to obtain a 0.3 second transmit interval t_{int} ($\approx 3 \text{ Hz}$) as shown by Equation 18:

$$t_{int} = 0.3s * 10\% = 30ms \quad (18)$$

The required frame size can then be approximated:

$$L_{frame} = 4000B/s \cdot 0.03s \approx 120 \text{ bytes} \quad (19)$$

These calculated parameters can then be used to place the Si4468 radio module in the desired operating state. This is achieved through the official Si4468 API which contains various API calls to configure the state of the radio transceiver. The resulting transceiver is used in either CW mode for single-tone testing or 2FSK mode with the above-mentioned parameters when transmitting data packets.

3.1.3 Tx/Rx Configuration

Radio transceivers typically have a split transmit (Tx) and receive (Rx) path, where the RF path to the antenna is selected/controlled by a digital RF switch. We require around 26 dB or greater of isolation (given a maximum of 36 dBm of TX power) between the Rx and Tx paths of the RF switch since the Si4468 transceiver has a maximum Rx input power of +10 dBm, as per the datasheet. It should be noted that, for the purposes of this dissertation, the transmit functionality (Tx) is considered the most important for actual output power control, with the receive (Rx) path being the subject of future work.

The SKY13290-313LF [82] 10W SPDT RF switch by Skyworks Solutions provides an isolation greater than 30 dB at 500 MHz as well as a input 0.1dB compression point ($IP_{0.1dB}$) of around 37.6 dBm and is used to switch between the Tx and Rx paths of the transceiver. In order to protect the transceiver's, a 10 dB attenuator is also added in the Rx path to further increase the isolation at the receiver (Rx) input.

The attenuator should be removed in a final design, however, as it reduces the sensitivity of the transceiver and negates the effect of the PA Tx stage. The input 0.1 dB compression point for an RF switch represents the input power at which the insertion loss of the RF switch increases by 0.1 dB compared to its linear behavior. It is generally recommended to operate with an input power below ($IP_{0.1dB}$).

3.1.4 Block Diagram

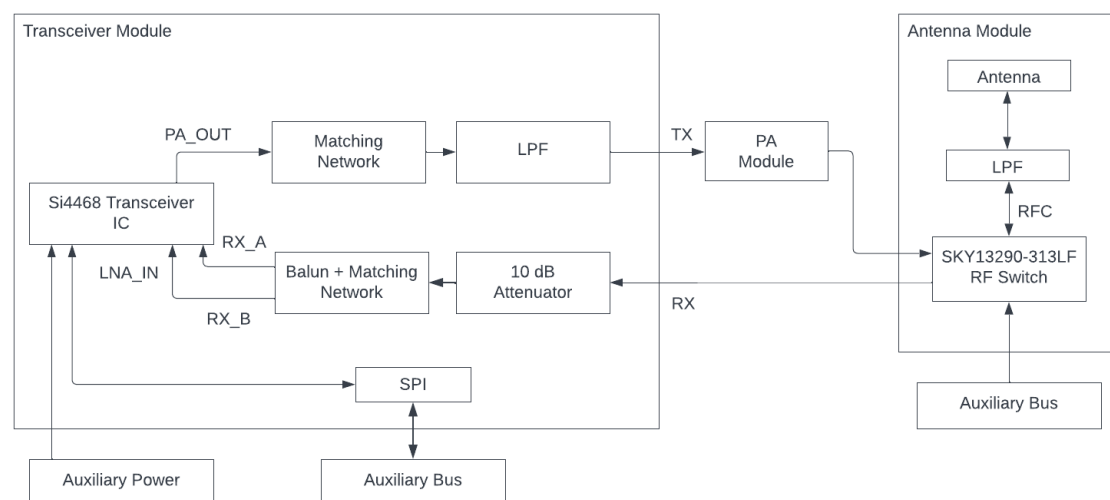


Figure 3.2: High-level block diagram of transceiver module, including all the matching and filtering networks. The antenna module is shown as a separate, theoretical block.

Figure 3.2 shows the resulting block diagram of the transceiver module. The module contains a Tx output port and Rx input port as well as a connection to the auxiliary power and the auxiliary data bus.

The Si4468 transceiver is a core component of the module and handles CW and 2FSK transmit (Tx) events sent to the transceiver via SPI by a 8-bit microcontroller. Additionally, there are two internal RF paths in the Si4468 transceiver. The first is the internal class-E power amplifier (PA) output which is fed into an external matching network responsible for presenting the optimum load impedance to the class-E PA.

The output of the matching network is then passed through an LC low pass filter (LPF) stage to reduce the power of harmonic tones. In addition, the receive path of the Si4468 contains an LNA (low-noise amplifier) with a differential input. The RF signal received at the monopole antenna, however, is single-ended and so a balun is required in the receive path to perform the single-ended to differential signal conversion.

A matching network is also required to present the optimum load impedance looking in to the LNA path for the lowest noise figure (NF). The balun and matching network can typically be combined into a single circuit. The antenna module is also shown in Figure 3.2 and contains an input (Tx) and output (Rx) port. As mentioned, both the Tx and Rx ports are multiplexed via a SPDT (Single Pole Double Throw) RF switch. Finally, the RF switch RFC port is connected to a final stage LC low pass filter (LPF) whose output is attached to a $50\ \Omega$ load such as an antenna.

3.2 Power Amplifier Module

3.2.1 Overview

The power amplifier module is the primary subsystem module responsible for sampling and controlling the RF output power to achieve a user defined set point in software.

It can be subdivided into the following subcircuits:

- Input drive/attenuation controller.
- Gate and drain bias controllers.
- Output power sampler.
- Drain current monitor.

3.2.2 Specifications

As an SRD device, we are limited to an ERP transmit power of 500mW (Figure 2.5). ERP transmit power (Equation 14) depends on the transmitter power itself as well as the maximum gain of the antenna relative to a half-wave dipole. The goal of the PA module is to achieve fine output power control (around 0.5 dB steps) using the two methods covered in the literature review, which are variable drain and input attenuation control.

A target resolution of approximately 0.5 dB allows us to measure power steps using an inexpensive spectrum analyzer (high noise floor, lower dynamic range) while bringing the cost of the required components down. A resolution lower than 0.5 dB would typically require a more accurate RF power detector, ADC, microcontroller, as well as a higher end spectrum

analyzer, and so a design limitation exists. The selection of a suitable RF power transistor is also an important step in the design of an RF power amplifier. Since we are operating in the VHF band, our transistor technology of choice will be LDMOS since these devices cost considerably less than GaN (Gallium Nitride) transistors, as mentioned in the literature review.

We would also like to select a power transistor with a non-linear simulation model available in order to perform load-pull and harmonic balance simulations to determine the optimum input and output impedance of the RF power transistor, harmonic tone power, and plot the output power vs drain bias and input power curves. Typically optimum source and load impedance data is provided in an RF power transistor's datasheet by the manufacturer but these impedance values are optimized for a particular application circuit which we do not reference. Hence, we will not use the predetermined load-pull data supplied in the datasheet.

Since our target output power is 1W (30 dBm) we select a transistor with a P1dB compression point of around 36 dBm in order to achieve good efficiency at maximum output power. It is usually a good idea to provide at least 3-6 dB of compression headroom or power 'back off' in order to maintain good linearity across the full range of output power. The AFT05MS004 LDMOS power transistor by NXP [83] is an inexpensive 4W (P1dB = 36 dBm) RF power transistor that meets all our requirements at a low price point. It also has a simulation model available for download on the NXP web product page.

The PA itself will operate in single-ended class-AB mode, which is a compromise between class-A and class-B operation to achieve higher efficiency and reduce component count. Theoretical efficiencies of around 50 % to 75 % are typically possible [84] in this mode of operation. Additionally, the AFT05MS004 has a power gain of approximately 20 dB at a drain efficiency of 75 % (using the narrow-band reference data from the datasheet). The power gain and efficiency can vary significantly depending on the chosen operating point, but the data provided serve as a working guide.

3.2.3 Block Diagram

Figure 3.3 shows the power amplifier (PA) block diagram. The PA module has a Tx input and Tx output port. The PA Tx input is connected to the Tx output of the transceiver module (Figure 3.2) while the Tx output of the PA module is connected to the Tx input of the antenna module. Digital integrated circuits such as the MCP3202 ADC [85] and MCP4726 DAC [86] are connected to the auxiliary bus and auxiliary power (3.3V).

Finally, the variable drain bias output of the power supply module (PSM) is connected to the drain bias input of the PA module.

The PA module also contains an input matching network that matches the output of the 30 dB digital attenuator to the input of the AFT05MS004 LDMOS power transistor, as well as an output matching network and low pass filter (LPF) stage, which provides the optimum load impedance to the AFT05MS004 MOSFET and reduces harmonic tone power.

The relevant subcircuits of Figure 3.3 will now be described in more detail.

3.2.5 Directional Power Sampling

In order to control the biasing or input drive to the PA, for both power control strategies, the RF RMS output power of the PA needs to be sampled and digitized. The value of RF output power can then be compared against the user set point in firmware before adjustments to the bias or input attenuator are made through a control algorithm.

A directional coupler is therefore required to couple a portion of the signal traveling towards the antenna or dummy load to the RMS logarithmic power sensor. In addition, an RF power detector and an ADC stage for the RF forward path are required to measure the average RF RMS power of the coupled signal and digitize the DC output of the RF RMS power detector.

The choice of directional coupler is mainly determined by the frequency of operation, coupling factor, insertion loss, as well as power handling capability. The coupling factor is not too important for the design, but must be chosen so that the RF power at the coupled port is reduced reasonably before being fed to the RF power detector. The Anaren 11302 coupler [88] is a 20 dB bi-directional coupler with a directivity of around -25 dB at 169.4 MHz, power handling capability of 100W and a maximum insertion loss of around 0.3 dB.

When selecting a suitable RF RMS power sensor, the choice is mainly driven by the available dynamic range in dB. Ideally, we would like a dynamic range of 40 dB or more (to cover the full 0 to 30 dBm output power range). The LTC5582 [89] power detector has a dynamic range of 57 dB and a logarithmic slope (K_{SLOPE}) of approximately 29.5mV/dB. A change of 0.5dB therefore corresponds to a 14.75mV change in the output voltage. We therefore need to choose an ADC with enough resolution to detect this small change in voltage. The voltage resolution of a slow speed, low-resolution ADC can be approximated by:

$$V_{RES} = \frac{V_{cc}}{2^n} \quad (20)$$

Here, n is the number of bits of resolution of the ADC and $V_{cc} = 3.3V$. Typical values of n are 8, 10, 12, 16 and 24 with each successive increase in resolution adding to the cost of the ADC. A 10-bit ADC has a voltage resolution of approximately 3.22 mV/step while a 12-bit ADC provides a voltage resolution of approximately 0.8 mV/step. At this point using a higher resolution ADC becomes impractical due to the cost and a 12-bit ADC returns more than 15 counts for a 0.5dB power step, which is a good starting point.

The MCP3202 [85] is an inexpensive 12-bit ADC with an SPI interface. It provides a fast data transfer rate so that a large number of power output samples can be obtained before any changes to the bias or input drive conditions occur. It should be noted that the output voltage vs input power transfer function of the LTC5582 power detector is not completely linear across its entire dynamic range. The mean output voltage vs RF input power graph can be found in the device datasheet. From the datasheet the power detector responds most linearly to an input power range of -50 dBm to -10 dBm.

ADC inputs and DAC outputs are also inherently noisy, and therefore some form of sample averaging or voltage smoothing using a shunt capacitor (typically 10 μF) must be utilized for each input / output channel, and this was the approach adopted for VARPA.

We know that our maximum RF output power from the AFT05MS004 is approximately 36 dBm and that the 11302-20+ directional coupler has a coupling factor of 20 dB. Therefore, at maximum transmit power the RF power at the coupled port will be around 16 dBm. Therefore, a 30 dB fixed attenuator is required between the directional coupler and the input of the RF power detector sensor to ensure good linearity throughout the maximum output power range.

It should also be mentioned that the power sampling mechanism uses a 50 Ω ceramic resistor which can be switched on and off using a SPDT RF relay (HF301 [90]). The HF301 provides good isolation (greater than 80 dB) at 169.4 MHz. During the power sampling and adjustment procedure, the 50 ohm load is switched in to allow RF power adjustment without RF radiation or an attached antenna.

3.2.6 Drain Current and Temperature Sense

In order to set the desired Q-point of the AFT05MS004 LDMOS power transistor, the drain current of the transistor needs to be continuously monitored. The voltage at the gate of the RF power transistor can then be ramped up until the drain current is approximately equal to 100 mA at which point the RF input to the PA can be applied.

To prevent thermal runaway of the RF MOSFET, both the drain current and temperature of the AFT05MS004 LDMOS power transistor must be monitored. The drain current sense circuit incorporates a current sense resistor, operational amplifier, and an ADC that converts the output voltage of the operational amplifier into a digital value which can then be converted into an equivalent DC current in firmware.

In order to conduct heat efficiently and obtain an accurate transistor temperature reading, the temperature sensor must make physical contact with the AFT05MS004 LDMOS power transistor. The datasheet for the AFT05MS004 indicates that the transistor has a heatsink tab connected to its source terminal. If we use a PTC thermistor as part of a voltage-divider network, then one side of the thermistor can be soldered to the grounded heat-sink tab of the power transistor. The output voltage of the network can then be digitized using an ADC and the actual device temperature calculated in software.

Power reflection when no load (antenna) is attached during transmit (Tx) is also another issue that can result in significant power dissipation in the RF power transistor. A suitable transistor must be chosen that can handle a high VSWR mismatch with no load connected for an extended period of time without failure.

3.3 Power Supply Module

3.3.1 Overview

The primary goal of the power supply module is to provide an energy source to all the system modules that are ultimately used to generate the required RF output tone. This includes a universal 3.3V bus as well as a voltage regulation circuitry for the variable drain bias controller that is required.

The Si4468 transceiver [67] by Silicon Labs used in the transceiver module requires a 3.3V supply and a transmit quiescent current $I_{Q,TX}$ greater than 108 mA as per the device datasheet

for around +20 dBm of RF output power using a class-E matching scheme. Additionally, the PA module makes use of the AFT05MS004N 4W LDMOS power transistor [83] from NXP which requires a 7.5V supply to generate 4W (≈ 36 dBm) of RF power for a narrow-band (high Q) matching configuration at a drain efficiency (η) of 75 %.

Furthermore, the required operating current at 36 dBm of RF output power is not specified and must be approximated for our class-AB PA assuming that we do not operate too far from P1dB compression. However, we will assume wideband operating conditions (at medium to low Q) for our design and therefore $\eta = 0.61$ (61 %) according to the transistor's datasheet.

Rearranging Equation 10 for $I_{Q,TX}$:

$$I_{Q,TX} = \frac{P_{out}}{\eta V_{DC}} \quad (21)$$

Since $V_{DC} = 7.5V$, $\eta = 0.61$ and $P_{out} = 4W$, using equation 21 we get $I_{Q,TX} \approx 0.9A$. Therefore, the RF power control mechanism of the system requires that the power supply module generate a variable drain or supply bias between 0V and 7.5V to output 4W of RF power. As stated above, our requirement was to generate at least 1W of RF power. It should be mentioned that the maximum RF output power of the PA will be restricted in firmware rather than in the circuit itself. Therefore, we simply assume a 4W output power capability for the PA.

A linear voltage regulation circuit consisting of a N-channel MOSFET and an operational amplifier in feedback mode (Figure 2.4) will be implemented to meet this requirement. An alternative solution involves utilizing an off the shelf LDO (low drop out) voltage regulator and a DAC. Variable LDO's typically have a feedback pin (FB) which can be manipulated using a DAC in a configuration known as a power supply margining circuit [91].

The issue with this approach is that it is generally difficult to design a power margining circuit with an output voltage less than the LDO reference voltage V_{REF} , which is typically around 1.5V with a single rail supply. A voltage output of less than 1.5V would be required to achieve an RF output power less than approximately 20 dBm according to simulation. Finally, the PA drain bias source should be able to be turned off and on via a digital control signal from the auxiliary or control module.

A simple solution to this requirement is a high-side switching circuit that consists of a P-channel MOSFET and a NPN transistor. The use of a linear regulator has the advantage of being low-cost and simple to control, but suffers from lower efficiency than a dedicated switch-mode buck/boost regulator.

3.3.2 Specifications

With a high-level overview for the power supply module completed, a list of specifications can be derived. The DC drain current required during transmit ($I_{Q,TX}$) is used as a starting point. A design factor of 2 can be applied to the required DC drain current $I_{Q,TX}$ during transmission to provide operational headroom and to account for additional components, future design requirements, and uncertainties in design calculations. The power supply module should therefore provide a variable 7.5V rail with a minimum current handling capability of $\approx 2A$.

Linear regulators have the advantage of producing less electrical noise at the output port but as mentioned suffer from large internal power dissipation when the input-output voltage differential of the regulator becomes large for a given drain current (ohm's law). It is common to use heatsinks to manage the thermal stability of the RF power transistor in this type of application.

The power dissipated for a continuous Tx tone (100 % duty cycle) in the N-channel MOSFET based linear regulator for our PA can be linearly approximated by Equation 22 (using V_{DC} and $I_{Q,TX}$ to calculate the equivalent linear resistance at the drain) assuming we are operating at P1dB compression:

$$P_{diss} \approx (V_{in} - V_{out}) \cdot V_{out} \cdot \frac{I_{Q,TX}}{V_{DC}} \quad (22)$$

Substituting and setting $V_{in} = 12V$ and setting the first derivative equal to 0:

$$\frac{d}{dV_{out}} P_{diss} = 12 - 2 \cdot V_{out} = 0, V_{out} = 6 \quad (23)$$

At $V_{out} = 6V$ we obtain the maximum power dissipation in the MOSFET ($P_{diss} \approx 4.3W$).

Note, however, our actual transmit duty cycle is set to (1-10) %, as per EU SRD regulations, which significantly reduces power dissipation in the N-channel MOSFET to around 0.432W assuming that the thermal time constant of the heatsink τ_s is large compared to the switching period. However, P_{diss} is too large to select a SMD mounted MOSFET. Rather, a through hole MOSFET with a dedicated heatsink must be used instead in order to not thermally stress the power MOSFET.

To calculate the required thermal resistance (K/W) of the heatsink (R_{sa}) we can use Equation 24, where T_j is the desired junction temperature, T_a is the ambient temperature, and R_{jc} and R_{cs} are the thermal resistances of the MOSFET's junction-to-case and case-to-sink junctions:

$$R_{sa} = \frac{T_j - T_a}{P_{diss}} - R_{jc} - R_{cs} \quad (24)$$

The thermal resistance of the selected heatsink must be equal to or less than R_{sa} .

Switching regulators have the advantage of being more power efficient but generate switching noise which makes them unsuitable for low-noise RF applications. Switching noise directly translates into AM modulated sidebands on the output which is undesirable. With this in mind, a specific N and P channel MOSFET and operational amplifier need to be selected based on the identified requirements. However, there are a large number of commercial variants for both components on the market.

The choice of MOSFETs in this case is primarily determined by the maximum drain to source breakdown voltage V_{DS} , gate threshold voltage $V_{GS(th)}$ and continuous drain current I_D while the choice of the operational amplifier is less critical and is mainly determined by how close the output voltage can swing relative to the power rails. For all components, the availability of simulation models is also important.

We would ideally like a V_{DS} greater than about 2 times the required 7.5V rail and a I_D larger than 2A as identified earlier. First, a suitable MOSFET must be selected for the drain bias voltage regulation circuit. The IRLB3813 [92] by International Rectifier is a N-channel power MOSFET with a maximum V_{DS} and $V_{GS(th)}$ of 30V and 2.35V respectively, I_D higher than 2A and a maximum operating junction temperature (T_j) of 175°C.

Using the thermal resistances provided by the IRLB3813's datasheet and a target junction temperature of $T_j = 105^\circ C$ we can substitute into Equation 24 to yield $R_{sa} \approx 17.3$ deg C/W. Therefore, we require a heatsink with a thermal resistance (R_{sa}) lower than or equal to 17.3 deg C/W. The E2A-T220-25E [93] is a heatsink by Ohmite with a thermal resistance of 16.4 deg C/W. The choice of the P-channel MOSFET for the high-side switch is not as important and is mainly centered around the maximum rated V_{GS} , V_{DS} and $R_{DS(ON)}$ of the device. The IRF7205 [94] is a P-channel power MOSFET with a $R_{DS(ON)}$ of around 0.07 Ω and a maximum power dissipation of 2.5 W.

Next, the LT1490A [95] by Linear Technologies, a rail-to-rail operational amplifier that supports single-rail operation, is selected for the linear regulation circuit and should be adequate for our design requirements, as this is a high performance operation amplifier with a low offset voltage error ($< 0.5\%$).

Finally, an LDO with a fixed output needs to be selected for the 3.3V rail which is responsible for powering all the subsystem electronics except the LDMOS PA MOSFET. The MIC2920A-3.3 [96] by Microchip Technology is a low-cost, high-accuracy, 400mA LDO with built-in thermal overload protection. Of primary importance is the maximum current capability in this case, which is more than the required 108 mA.

It should be mentioned that the P-channel high side MOSFET will be used to realize the 10% duty cycle requirement for the system. A switching frequency of 3.33 Hz will be used and is derived in more detail in the subsequent detailed design section.

3.3.3 Block Diagram

The resulting subsystem block diagram for the power supply module (PSM) is shown in Figure 3.4. The inputs/outputs to the PSM include the 12V DC PSU, variable drain bias output, auxiliary power output, and auxiliary bus. The drain bias regulator circuit is driven by the MCP4726 DAC using a voltage feedback configuration to achieve a variable 0 to 7.5V drain bias voltage.

The auxiliary module communicates with the MCP4726 DAC over the auxiliary bus (I2C, SPI) and sets the specific DAC output voltage that is required to achieve a specific drain bias output voltage. In addition, the power supply module contains a P-channel MOSFET high-side switch which enables or disables the drain bias output. Power supply decoupling capacitors are also included in the design to filter out low and high frequency noise that is coupled into the power supply circuit by the environment.

Since this is effectively a linear regulator, the noise performance of the power supply module will be better than that of a high-frequency switch-mode supply. The high thermal dissipation of the linear regulator is also overcome by using a low duty cycle ($< 10\%$) for transmit, which is required to meet the SRD (short-range device) regulations. The limitation of using a linear regulator as apposed to using a real switching regulator is, again, reduced energy efficiency due to the absence

of an inductive element to store and release energy, with the linear regulator dissipating energy as heat, as mentioned.

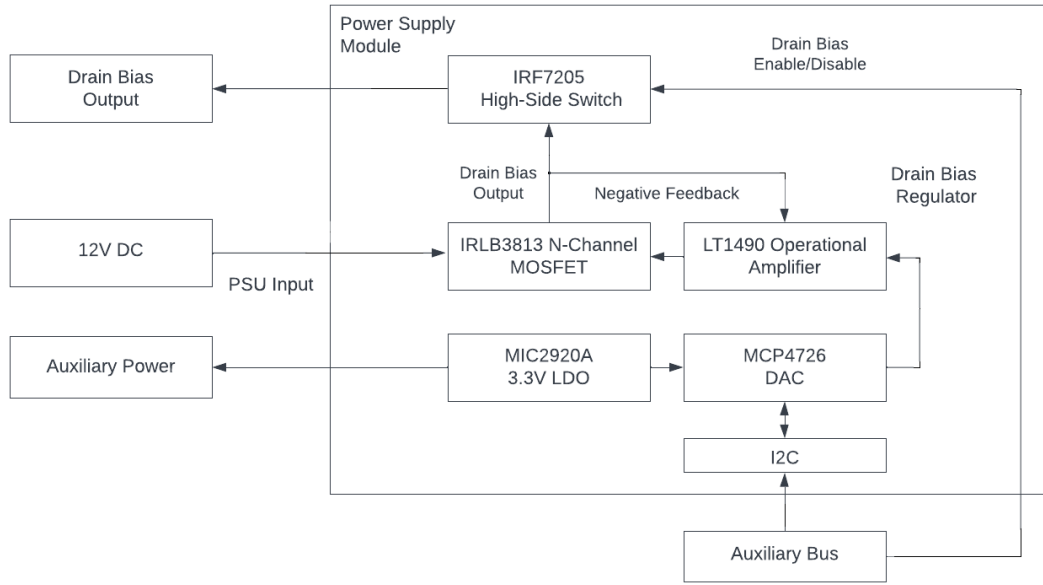


Figure 3.4: Detailed power supply block diagram, with the IRLB3813 MOSFET used in a negative feedback configuration.

3.4 System Controller Module

3.4.1 Overview

The auxiliary control module is the primary digital coordinator of the system. This module is ultimately responsible for controlling the various DAC and ADC integrated circuits, the F1956NBGI8 digital attenuator, as well as handling SPI and I2C communication between the system modules. The auxiliary module is also responsible for controlling the RF switch in the RF power amplifier and transceiver modules. Additionally, the auxiliary control module consists primarily of an 8-bit microcontroller along with the required passive circuitry to enable freestanding operation.

3.4.2 Specifications

Since there are 2 high-speed SPI ADCs across all modules, 5 pins will be required to operate these devices, 3 of which are the SCK, MOSI and MISO lines, with the other 2 corresponding to the CS pins.

However, the F1956NBGI8 digital attenuator implements a 3-wire half-duplex SPI interface and uses the MOSI pin to set the attenuation value. An additional pin for the chip select (CS) will be required, increasing the total required SPI pin count to 6. Low-speed I2C peripherals, on the other hand, only require a SCL (serial clock) pin and an SDA (serial data) pin and use an address to identify a particular slave. Therefore, an additional 2 pins will be required for the I2C devices.

For the SKY13290-313LF RF switch, 2 digital pins per switch are required to control the state of the switch. An additional pin is also required for the HF301 switch. Therefore, an additional 3 IO pins will be required. This brings the total number of digital I/O pins required to 11. We

therefore require a microcontroller with at least 12 digital pins (but ideally more for future design changes). The bus width is not too important, and generally an 8-bit microcontroller is more than sufficient for control applications compared to more expensive 32-bit variants.

The ATmega328p [97] (Figure 3.5) is a popular choice due to its large pin count, hardware support for SPI and I2C, low cost, and ease of programming. In addition, the ATmega328p has 32 KB of flash memory, a 1 KB EEPROM, 24 IO lines, and 2 KB of SRAM all at an affordable price, and both SMD and DIP variants of the IC exist, which makes prototyping easier.

All firmware for the VARPA project is written from scratch in C and compiled with avr-gcc using a GNU makefile. The firmware for ATmega328p can be flashed using an ICSP (In Circuit Serial Programming) compatible programmer. The use of an external 16 MHz quartz crystal is also recommended for improved clock stability. Various scripts will be developed in Python to generate constants used by the firmware (such as the RMS power detector calibration data). All code developed for VARPA can be found via a link in the [Appendix](#) of this dissertation.

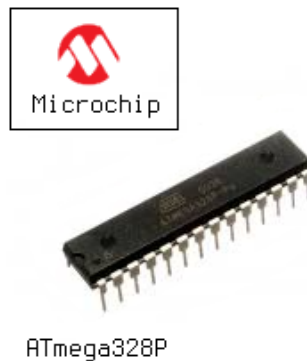


Figure 3.5: 8-bit ATmega328p microcontroller, which is used as the main controller for the system.

3.5 Conclusion

Through a high-level design analysis of the system, four core subsystem modules were identified. Then block diagrams for 3 of the most important subsystems were created to serve as a guide for stage 4 of the methodology: the simulation and detailed design phase.

These are the:

- Power Supply Module (PSM) - provides power to the system.
- Transceiver (Tx) Module - generates the required CW source tone input.
- Power Amplifier Module (PAM) - responsible for regulating the RF output power.

With all the high-level details of each module specified, the design of the system can therefore proceed to the next phase, which is the detailed design and simulation stage. During this next stage, detailed schematics will be created for each system module and the values of network components calculated. Various simulations will also be performed to optimize the final design.

4 Detailed Design and Simulation

With the high-level subsystem block diagrams developed, the detailed design phase follows, which involves calculating component values for each subsystem circuit. These values are derived mainly through simulation and manual calculation using design equations. More specifically, schematics for all four system modules are developed. This includes the detailed design of the power supply module, transceiver input and output matching networks, LC harmonic filters as well as the power amplifier module .

The PA design includes a simulated load-pull and stability analysis, followed by the synthesis of all matching and filtering networks and, finally, a sensitivity analysis is performed to ensure that component tolerances do not significantly effect PA performance.

4.1 Power Supply Module

4.1.1 Drain Bias Control Circuit

Figure 4.1 shows the schematic of the analogue drain bias control circuit. As indicated, the schematic consists of a high-side switching circuit as well as a linear voltage regulation circuit.

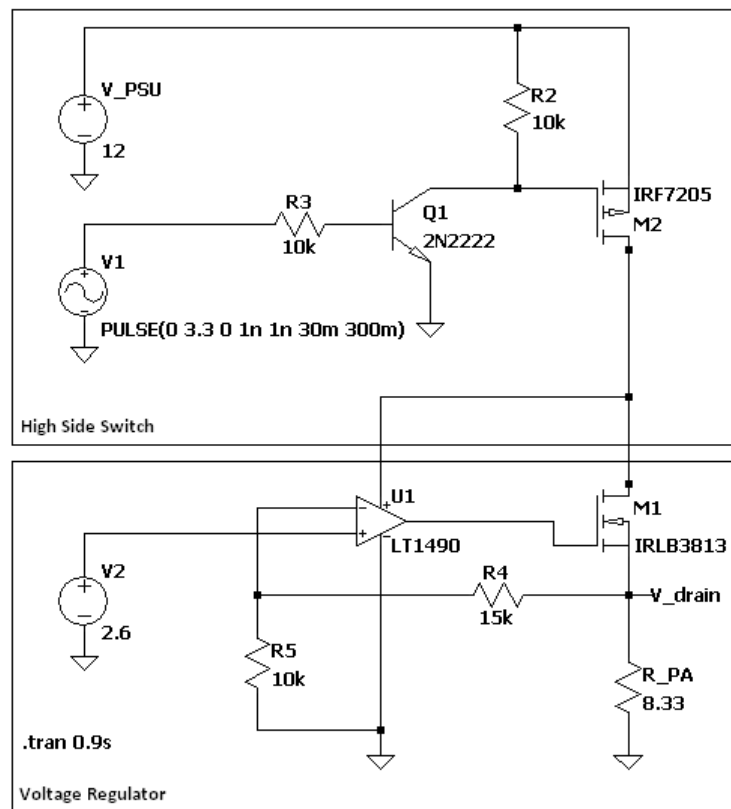


Figure 4.1: Drain bias control circuit which is used to supply the variable 0 to 7.5V drain supply voltage to the RF power amplifier module.

V_PSU represents the input from the 12V DC PSU which powers the system, while **V1** and **V2** are voltage control inputs that change the state of the power supply module. When **V1** is set to 3.3V or 0V, the high side switch is enabled or disabled which can be used to enable/disable the drain bias output to the PA module.

The high side switching circuit consists of two resistors (**R3** and **R2**), a NPN transistor (**Q1**) as well as a P-channel MOSFET (**M2**). **R2** serves as a pull-up resistor so that the gate of **M2** is at the same potential as the source by default, therefore disabling **M2** until the **V1** control signal is set to 3.3V.

Q1 is used to pull the gate of **M2** to ground when **V1** is applied at the terminal of **R3**, which limits the current through the base of **Q1**. The values of **R2** and **R3** are chosen so that no excessive quiescent current flows. 10 k Ω is a common value used for logic circuits. **R_PA** represents the equivalent DC resistance of the PA module ($\approx 8.33\Omega$) at 4.9W of RF output power.

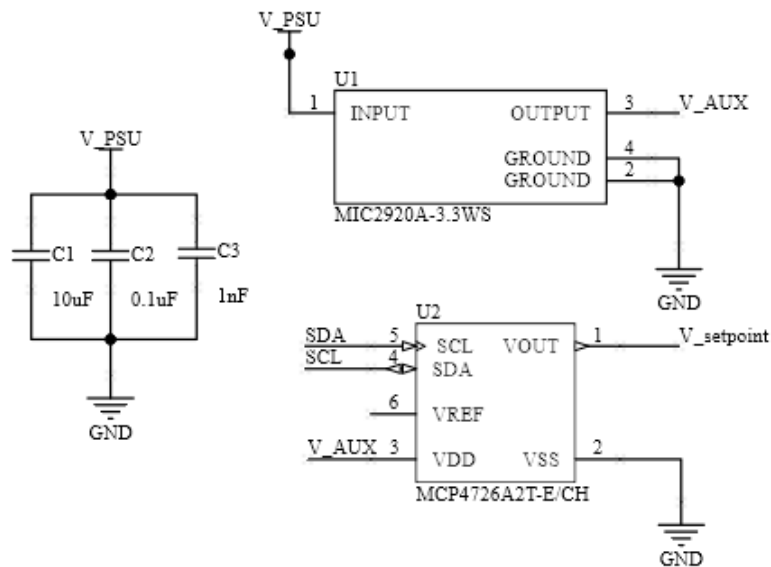


Figure 4.2: Schematic of the rest of the power supply module, with the 3.3V linear regulator, DAC and AC decoupling capacitors shown.

The remaining section of the power supply module is shown in Figure 4.2. It consists of two integrated circuits (**U1** and **U2**). **U1** is the 3.3V low drop-out regulator which supplies auxiliary power (**V_AUX**) to the rest of the system.

U2 is the MCP4726 DAC which is used to set the input voltage (**V_setpoint**) of **U1** in Figure 4.1. **VREF** is left unconnected as the MCP4726's internal voltage reference circuit is used instead. **SDA** and **SCL** are the I2C data lines and are connected to the auxiliary module. Finally **C1**, **C2** and **C3** are the power supply decoupling capacitors that are used to suppress low-frequency and high-frequency noise. The values of these capacitors are not too critical; however, a suitable range of values must be selected so that the SRF (self-resonant frequency) of the capacitors is well above the intended decoupling frequency. Values of 1nF, 0.1 μ F and 10 μ F are common in bypass applications.

4.1.2 Principle of Operation

U1 and **M1** of Figure 4.1 are used to control the drain bias output voltage (**V_drain**). The principle of operation is quite straightforward. **U1** is set in negative feedback mode and samples the output of the source terminal of **M1**.

M1 acts as a voltage-controlled current source, so the voltage **V_{drain}** depends on the current through **R_{PA}** which is set by the gate voltage of **M1**. The feedback gain, given by Equation 25 is set by **R4** and **R5** and is equal to 2.5 for the circuit shown in Figure 4.1.

$$A_v = 1 + \frac{R4}{R5} \quad (25)$$

This allows a range of **V_{drain}** of 0 to 8.25V, with a **V_{setpoint}** provided by **U2** of Figure 4.2 of 0 to 3.3V. It should be noted that the gate threshold voltage ($V_{GS(th)}$) of **M1** (IRF7835) has a maximum value of 2.35V as per the device datasheet. This means that the output of **U1** is 2.35V higher than $(2.5 \cdot V_{setpoint})$ in the worst case. This limits the maximum theoretical output voltage of **V_{drain}** to around 9.65V. Since our maximum output power is 4W (36 dBm) we require a maximum **V_{drain}** of 7.5V.

Figure 4.3 shows the result of a transient simulation of the schematic in Figure 4.1 with a **V_{drain}** of 6V (resulting in maximum power dissipation). The high side switching frequency and duty cycle are set to 3.33 Hz and 10 % respectively which is equivalent to the time on air for our transmit stage. It can be seen that an average power dissipation of ≈ 0.43 W is obtained or ≈ 4.3 W at 100 % duty cycle.

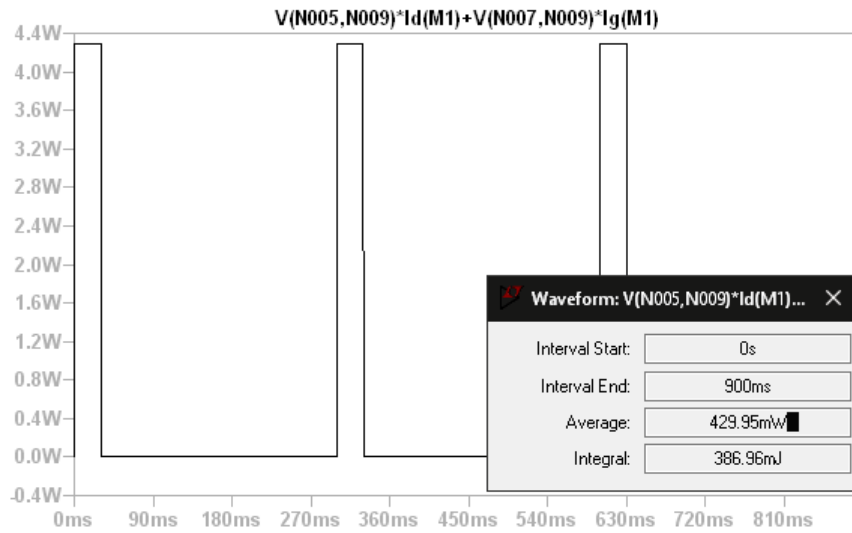


Figure 4.3: IRLB8743 transmit power dissipation with simulated input control waveform.

4.2 Transceiver Module

4.2.1 Tx Class-E PA Network

The proposed LC network that should be attached to the class-E power amplifier output of the Si4468 transceiver is shown in Figure 4.4. It is seen to consist of three separate stages. The first stage is a bias-tee section which isolates the DC and RF paths which is necessary to prevent RF from leaking into the DC supply.

At the same time, capacitor C1 blocks any DC component from reaching the output of the network, which prevents DC biasing issues at the output. This is followed by a two element LC matching network which presents the optimum load impedance to the class-E RF power amplifier output of the Si4468 transceiver. This is required to ensure that sufficient RF drive is

delivered to the input of the power amplifier (PA) module. The third stage consists of a $50\ \Omega$ fifth-order LC π -filter which further attenuates any remaining harmonic content before reaching the PA input. Each stage will now be described in more detail, including the calculations for deriving component values.

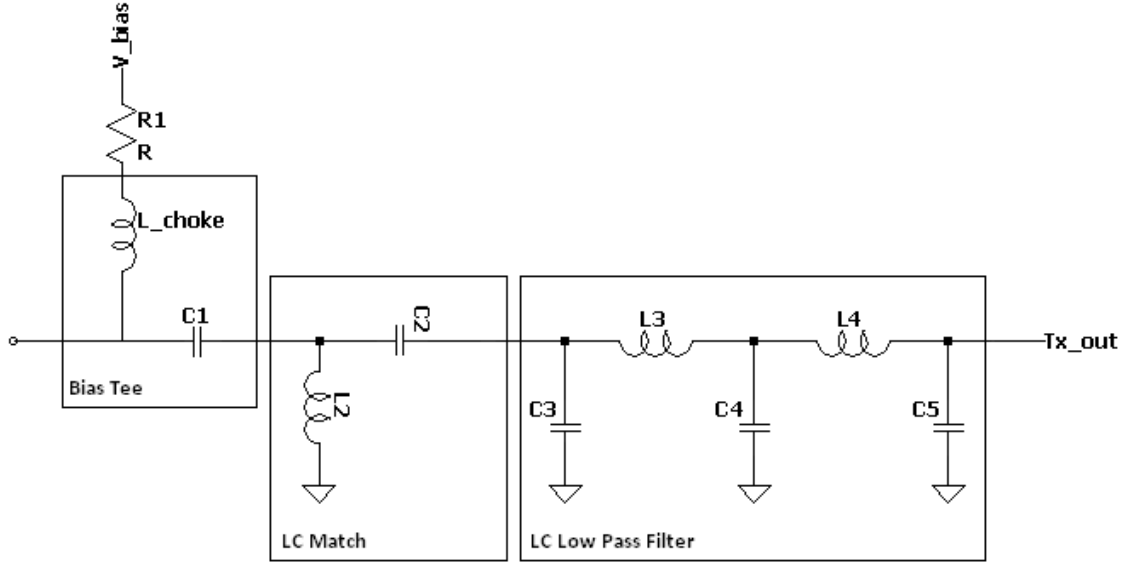


Figure 4.4: Transceiver (Tx) power amplifier output network, with the bias-tee, LC matching network as well as the 5-th order LC low pass π -filter shown.

4.2.2 Tx Class-E PA Matching Network

For the Si4468 RF transceiver, the optimal load impedance to present to the class-E RF power amplifier can be calculated using Equation 26.

$$Z_{opt} = \left(\frac{0.2815}{2\pi f_c C_{SHUNT}} \right) e^{j49.05^\circ} \quad (26)$$

The value of C_{SHUNT} for the Si4468 transceiver is provided in the application note AN648 by Silicon Labs [98]. Therefore, with $C_{SHUNT} \approx 2.5\text{pF}$ and $f_c = 169.4\text{MHz}$, Z_{opt} can be calculated using Equation 26.

Substituting into Equation 26 leads to a Z_{opt} equal to $(69.33 + 79.9j)\ \Omega$.

Let $Z_0 = 50\ \Omega$, $R_L = 69.33\ \Omega$, $X_L = 79.9j\ \Omega$

$L2$ of the LC matching network (Figure 4.4) can then be calculated analytically using Equations 27 and 28 [23]:

$$B = - \frac{R_L^2 + X_L^2}{X_L + \sqrt{\frac{R_L}{Z_0}} \sqrt{R_L^2 + X_L^2 - Z_0 R_L}} \quad (27)$$

$$L2 = -\frac{B}{2\pi f_c} \quad (28)$$

Substituting into both equations leads to an **L2** = 57.33 nH

C2 can then be calculated using Equations 29 and 39:

$$X = \frac{B(R_L - Z_0) - Z_0 X_L}{-R_L} \quad (29)$$

$$C2 = \frac{1}{2\pi f_c X} \quad (30)$$

Again, substituting in both equations yields **C2** = 12.59 pF.

The impedance transformation using components L2 and C2 can also be visualized using a smith chart.

4.2.3 Tx PA Low Pass LC π -Filter

C3, L3, C4, L4 and **C5** of Figure 4.4 form an LC fifth-order, low pass π -filter that is used to further reduce the harmonic power of the Si4468 PA output stage.

There are numerous ways to design an RF filter. Traditionally RF filter design has been achieved using tables of polynomial constants which are chosen depending on the desired filter order and response. However, a more modern and flexible approach is to utilize filter design software such as MATLAB's 'RF Toolbox' which allows for more powerful design optimization.

The first step in the design of the LC π -filter is to choose the filter type, passband attenuation, order, and input and output impedance. Two common RF filter types are the Butterworth and Chebyshev filters. In our case, we wish to maximize the attenuation past the passband and so a steep roll-off is desirable. Therefore, we choose the Chebyshev filter type. A steeper stopband roll-off can theoretically be obtained with a larger passband ripple. However, the stopband ripple should not be too high such that component tolerances cause a change in the expected point of operation in the passband (point of lowest insertion loss).

The generally accepted rule of thumb is to limit the maximum passband ripple to around 0.5 dB. Next we need to select the filter order. In order to achieve around 40 dB of attenuation at the second-order harmonic, we require a fifth-order LC π -filter with an input and output impedance of 50 Ω .

A MATLAB script was developed to calculate the π -filter component values. The script uses a number of predefined design parameters and then runs an optimization algorithm to find the optimum passband frequency that results in a near-perfect 50 Ω match. Figure 4.5 is a plot of the filter response (S_{21}). It can be seen that the insertion loss at 169.4 MHz is negligible and that we have achieved our target attenuation of approximately 40 dB at the second harmonic (338.8 MHz).

We can compute the filter input impedance at 169.4 MHz using the following command in MATLAB:

```
gamma2z(gamma2z(main(final_filter.parameters(Fp*10^6)))
```

After running the MATLAB command above, we obtain an input impedance of $(50.3460 + 0.0910i) \Omega$ at 169.4 MHz for the LC π -Filter after optimization. The final component values of the LC filter as shown in Figure 4.4 are as follows:

C3, C5 = 30.5 pF and **C4** = 45.44 pF

L3, L4 = 54.97 nH

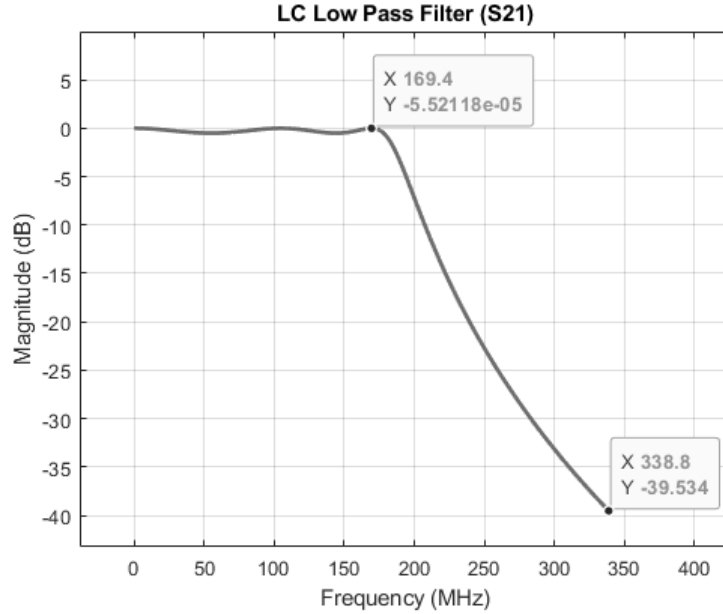


Figure 4.5: S_{21} of the Tx LC low pass π -filter, with markers at the point of lowest insertion loss and at the second harmonic tone.

It should be noted that a LC tee filter could also have been designed in place of the LC π -filter as the same S_{21} response would be obtained. The LC π -filter configuration was chosen in order to obtain more practical values of capacitance and inductance.

4.2.4 Tx Bias-Tee

The values for the remaining components: **R1**, L_{choke} and **C1** can now be calculated. The purpose of the RF choke (L_{choke}) is to present a high impedance to the RF signal so that it is redirected to the matching network while presenting a low impedance to DC. As a general guideline, when using RF inductors in filtering or matching applications, the SRF of the inductor should be at least 10 times the fundamental frequency. However, in choking applications the SRF is typically chosen to be at or near the frequency to be suppressed [99].

One rule of thumb is to choose an RF inductor with a SRF that is at least $1.5f_{choke}$, where f_{choke} represents the primary choking frequency (169.4 MHz). As a first step, we can calculate the input impedance of the matching network:

$$|Z_{opt}| = \sqrt{R_L^2 + X_L^2} \quad (31)$$

After substituting for R_L and X_L we obtain $|Z_{opt}| \approx 211.5 \Omega$. Our goal is to present an integer multiple (n) of this impedance to the PA output port at a frequency below the SRF of the inductor. Our target SRF in this case is $1.5f_c \approx 250$ MHz. This integer multiple (n) generally ranges from 1 to 10. We use $n = 5$ for this design.

L_{choke} can then be calculated using Equation 32:

$$L_{choke} = \frac{5 \cdot 211.5}{2\pi 169.4 \cdot 10^6} \approx 1 \mu H \quad (32)$$

A table of SRF frequency vs inductance was obtained online [100] for SMD 0805 size inductors which indicates that the SRF of a 1000 nH inductor of this package type is above 250 MHz (typical). Hence, our choice of a 1 μ H RF choke is suitable for this design.

To calculate the value of **C1**, the impedance of the coupling capacitor should be significantly less than $|Z_{opt}|$. Capacitors typically have a higher rated SRF compared to inductors, so a wider range of component values can be used in the design. The choice of the impedance divisor is not too important, but we would like to present a low-impedance path to the RF signal at the PA output port. For this particular design, we choose $n = 250$. The upper limit of n is determined by the SRF of the ceramic capacitor, which should be just above the coupling frequency.

$$C_1 = \frac{1}{2\pi 169.4 \cdot 10^6 \cdot \frac{211.5}{n}} \approx 1 \text{ nF} \quad (33)$$

4.2.5 Rx LNA Balun-Matching Network

The parallel equivalent resistance and capacitance of the differential LNA input have been characterized by Silicon Labs using differential S-parameter measurements. Application note AN643 [101] states that the parallel resistance of the LNA (R_{LNA}) at 169 MHz is equal to 536Ω and that the parallel capacitance (C_{LNA}) is equal to 0.98 pF. Since we would like to obtain the best Rx performance possible, we choose to design a four-element matching network and balun hybrid topology.

Since the design details of the Si4468 LNA input circuit are not as important as those of the PA output circuit, no detailed design analysis is provided. Figure 4.6 depicts the 4 element match-balun network that provides the functionality of a matching network and balun.

L1 of Figure 4.6 can be calculated using Equation 34:

$$L_1 = \frac{\sqrt{50 \cdot R_{LNA}}}{2\pi f_c} \quad (34)$$

Since:

$$f_c = 169.4 \text{ MHz}$$

$$R_{LNA} = 536 \Omega$$

we obtain an **L1** = 153.8 nH

C1 can then be calculated using Equation 35:

$$C_1 = \frac{1}{(2\pi f_c)^2 \cdot L_1} \quad (35)$$

Therefore **C1** = 5.74 pF.

C2 is defined to be $(2 \cdot C_1) = 11.48$ pF

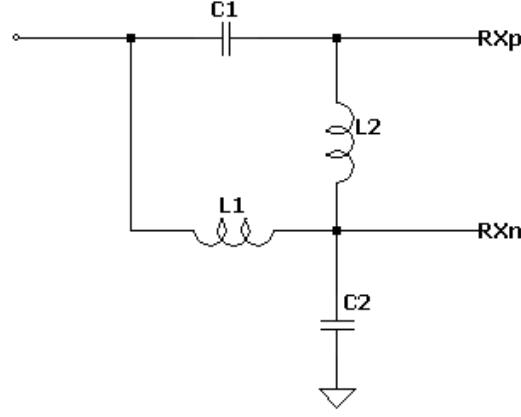


Figure 4.6: Integrated matching and balun network for the transceiver (Rx) LNA input.

Finally **L2** can be calculated. First two intermediate expressions (Equation 36 and 37) are defined:

$$L_{LNA} = \frac{1}{(2\pi f_c)^2 \cdot C_{LNA}} \quad (36)$$

$$L_M = 2 \cdot L_1 \quad (37)$$

These two expressions can then be used to calculate **L2** using Equation 38:

$$L_2 = \frac{L_{LNA} \cdot L_M}{L_{LNA} + L_M} \quad (38)$$

After substituting, **L2** = 229.29 nH.

The circuitry shown in Figure 4.6 is all that is required from a component point of view to enable receive (Rx) capability.

4.2.6 Final Stage Low Pass LC π -Filter

Harmonic content at the output of a RF PA is a combination of the harmonics generated by the PA itself as well as the harmonic content present at the input port. Therefore, in order to ensure an attenuation greater than 50 dB in the output path for the RF PA module, an optional LC low pass π -filtering stage should also be designed. It is worth mentioning that this additional filter stage is not included in the final prototype schematic and is intended to be used only if the 50 dB attenuation target is not met.

The exact order of the filter and hence required roll-off will be determined through a harmonic

balance (HB) simulation of the PA output stage. If it is found that the third-order filter does not attenuate the Tx signal, the filter order can be increased. However, as a first step, a filter order of $n = 3$ is chosen and designed using the same MATLAB script that was used to design the LC filter for the PA output path.

Figure 4.7 depicts the topology of the third order, LC low pass π -filter. After running the MATLAB script the following component values were obtained for a 169.4 MHz design frequency and 0.5 dB of passband ripple:

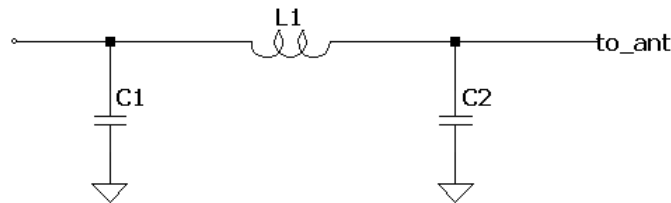


Figure 4.7: Final stage, LC low pass 3rd order π -filter for the transceiver (Tx) and RF power amplifier module.

$$C1, C2 = 25.99 \text{ pF}$$

$$L1 = 44.64 \text{ nH}$$

4.3 Power Amplifier Module

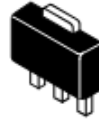
4.3.1 AFT05MS004 4W LDMOS Transistor

The AFT05MS004 LDMOS RF power transistor (Figure 4.8) is arguably the most important component of the PA module. It has a P1dB compression point of around 36 dBm, making it capable of delivering 4 watts of RF power to a matched 50Ω load. The device is not internally matched to 50Ω meaning that a load-pull simulation and input and output matching networks are required.

The datasheet of the device also lists both narrowband and broadband performance specifications. We are more interested in the narrowband or theoretical high-Q performance since we simulate individual LC components with ideal behavior. The narrowband performance has been characterized at 520 MHz with 36.9 dBm of output power.

The power gain achieved was around 20 dB at a drain efficiency of 74.9 %. This information was taken from the transistor's datasheet. Additionally, the device has a load mismatch VSWR ratio greater than 65:1 at all phase angles, meaning that we do not need to worry about causing damage to the device when no load is connected to the output of the power amplifier, which eliminates the need for additional RF protective circuitry such as isolators or circulators.

In order to design the input and output matching networks for the RF power transistor to achieve our target output power (30 dBm) and efficiency ($> 60 \%$) it is necessary to perform a series of RF design simulations.



SOT-89

Figure 4.8: The main RF LDMOS power MOSFET used in the power amplifier module is the AFT05MS004.

To this end, the publicly available non-linear model of the RF MOSFET was utilized. RF power amplifier design simulations are typically performed near the P1dB compression point to maintain good efficiency at maximum output power, since the transistor could technically be pushed into this region of operation.

4.3.2 DC I-V Simulation

The first step in the PA design process is to plot the I-V curves (drain current vs drain voltage) for different values of gate voltage. This allows us to determine the theoretical DC gate voltage (V_G) required to obtain the recommended drain quiescent current, which is stated as 100 mA in the transistor's datasheet for class-AB operation. V_G is required as it is used in subsequent design simulations.

However, it should be noted that the value of V_G obtained by simulation may not be the same as the gate voltage required in practice due to the tolerances of semiconductor manufacturing. Figure 4.9 depicts the DC simulation setup. There are two variable voltage sources (V1 and V2). V1 is set to range from the minimum value of $V_{GS(th)}$ (gate threshold voltage), which is listed as 2.5V in the datasheet, to 3.5 V (max value) in step sizes of 0.025V. V2 is set to vary from 0 to $2V_{CC}$, where V_{CC} is the supply voltage at the drain of the MOSFET. This is because the peak-to-peak voltage at the drain of a class-AB power amplifier with an RF choke reaches a theoretical limit of $2V_{CC}$.

The resulting DC I-V characteristic curves of the FT05MS004 RF MOSFET is shown in Figure 4.10. It was found that a gate voltage (V_G) of around 2.9V was required to obtain a 100 mA quiescent drain current (I_{DQ}). Additional gain parameters from Figure 4.10 could have been obtained by drawing a load line (such as maximum output power and efficiency). However, the additional information obtained using a load line is generally quite inaccurate, and so determining these parameters using separate simulations was the preferred approach.

It should be noted that we are operating the AFT05MS004 in class-AB mode, and therefore our conduction angle is not 360° but rather closer to 180° . By increasing V_G we operate in a more linear region with an increasing conduction angle and power gain. $V_G = 2.9V$ is therefore our gate bias or Q operating point.

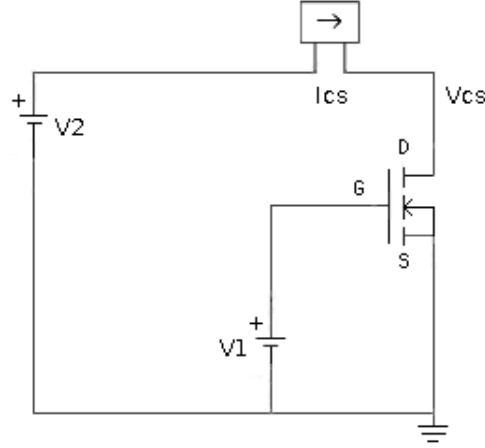


Figure 4.9: Simulation setup to plot the DC I-V curves of the AFT05MS004 RF power MOSFET.

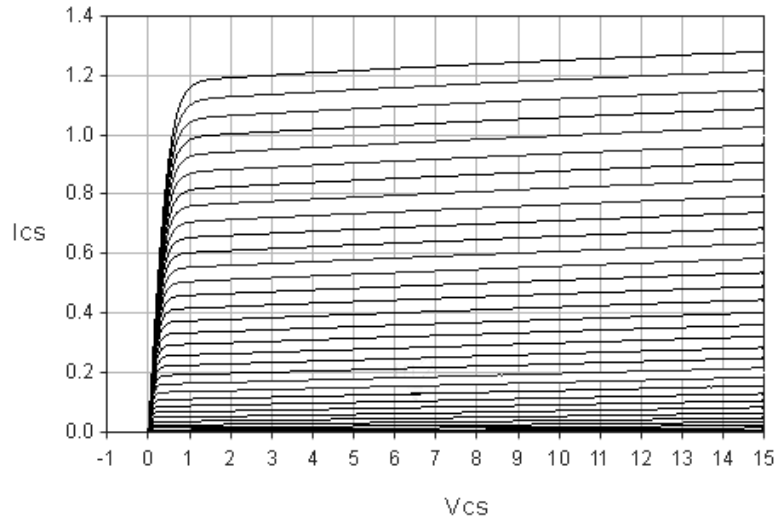


Figure 4.10: Resulting DC I-V curves for the AFT05MS004 RF power MOSFET.

4.3.3 Stability Analysis Simulation

The next step involves analyzing the intrinsic stability of the transistor using the Rollet (K factor) and the μ_{load} and μ_{source} stability factors mentioned in the literature review.

The simulation setup for stability analysis is shown in Figure 4.11. The main component of the simulation is the virtual NWA (network analyzer) which has two DC biasing ports (port 1 and port 2). The bias for port 1 is set to 2.9V as determined in the I-V simulation while the drain bias is set to 7.5V. The frequency range of interest is set to cover the entire VHF band (30 MHz to 300 MHz).

The series resistor (**R1**) shown in Figure 4.11 is added to stabilize the LDMOS device. The goal of the simulation is to determine the value of the stabilization resistor **R1**. With **R1** = 0 Ω , a K factor of 0.067 was obtained at 169.6 MHz and a μ_{load} of 0.708 indicating that our RF power transistor is unstable. The principle of the series feedback resistor is to dissipate energy in the feedback path such that Barkhausen's criterion for oscillation is avoided. There is a design

trade-off when choosing the value of the stabilization resistor, since increasing its value decreases the large signal power gain of the PA, which is undesirable.

Our Rollet (K) factor, as well as μ_{load} and μ_{source} should be greater than 1 around our target frequency f_c (169.4 MHz). The value of R (Figure 4.11) was then increased until the stability criteria mentioned above were met.

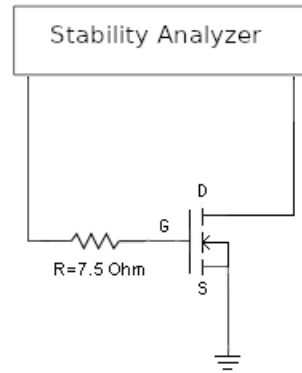


Figure 4.11: Simulation setup for the stability analysis of the AFT05MS004 RF power MOSFET.

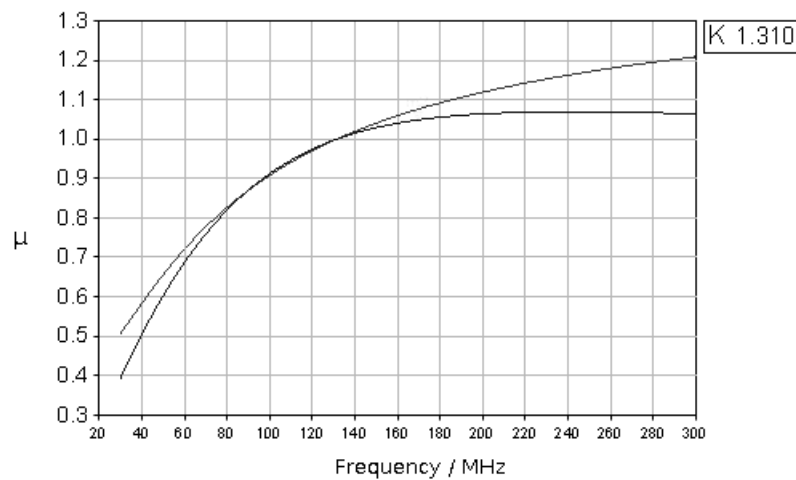


Figure 4.12: Results of the stability analysis simulation for $R_1 = 7.5 \Omega$. Value of K @ 169.4 MHz and plots of the geometric stability constants (μ_{load} , μ_{source}) for the RF power transistor up to 300 MHz.

Figure 4.12 is a plot of the stability constants for an $R_1 = 7.5 \Omega$. It can be seen that both the Rollet's stability factor and μ_{load} and μ_{source} are greater than 1 near the design frequency, hence our power transistor is now unconditionally stable. The stability analysis would then have to be performed again on the complete power amplifier circuit to ensure that the system remains stable after the addition of external LC network elements. If the system becomes unstable, then additional resistive elements can be added in series or parallel to dampen signal feedback as mentioned.

4.3.4 Load-Pull Simulation

Next, the optimum input and output impedances of the LDMOS power transistor need to be computed through a simulated load pull at an input power of +16 dBm (36 dBm output - 20 dB gain from the datasheet for narrow-band performance).

The simulation setup is shown in Figure 4.13. The primary component of the simulation is the 1-tone load-pull instrument which has two DC bias ports. The load-pull instrument accepts a wide range of different parameters including the gate and drain bias, center frequency, source power, etc. Initial values for the source and load impedance, which are required, can be obtained using the table of recommended source and load impedances for the narrow-band reference circuit in the transistor's datasheet.

From the RF transistor's datasheet it can be seen that at 170 MHz the optimum impedance of the source is equal to $(5.09 + 11.62j)\Omega$ while the optimum load impedance is equal to $(3.77 - 0.01j)\Omega$ for the reference circuit. These two values of the source and load impedance are used as the starting points for the load-pull simulation, with the input power set to +16 dBm. Figure 1 is a table with the resulting load-pull simulation data for the setup in Figure 4.13. The load-pull simulation setup shown in Figure 4.13 was configured to optimize for maximum PAE.

It can be seen that the output power delivered to a 50Ω load at the optimum input and output impedance is equal to 36.778 dBm at a PAE of 76.29 %. More specifically, these output conditions are met for a load impedance (Z_{load}) of $(5.01 + 3.65j)\Omega$ and a source impedance (Z_{in}) of $(7.78 - 11.26j)\Omega$

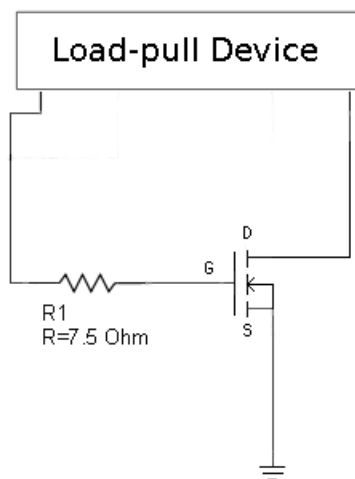


Figure 4.13: Simulation setup for a harmonic-balance based load-pull on the LDMOS RF power MOSFET.

Both the simulated load-pull data and the load and source impedance data found in the datasheet (for 170 MHz) are in close agreement, with the simulated load-pull data leading to more optimized results for our particular PA circuit. It can be seen in Table 1 that the DC power consumption at maximum power is ≈ 900 mA as calculated in the high-level design section of the power supply module.

Parameter	Value
Z_{in} / Ω	$7.78 - 11.26j$
Z_{load} / Ω	$5.01 + 3.65j$
PAE / %	76.29 %
Gain / dB	19.778
Drain Current / A	0.824
Output Power / dBm	36.778

Table 1: Results from the simulated load-pull for the LDMOS RF power MOSFET.

With these results our minimum target output power of 1W (30 dBm) has been achieved with +16 dBm of input power along with our initial PAE target of 60 % with no filtering networks attached. Our minimum large signal gain target of 10 dB has also been achieved, indicating that our LDMOS device choice was adequate for our initial design parameters. It should be noted that these values are theoretical and do not model component Q which would result in a lower large signal gain and PAE in practice.

4.3.5 Input Matching Network

With our load-pull simulation complete, the matching network synthesis can now take place. The design of the input matching network for the LDMOS device will be described in this section. We would like to match the 50Ω output of the digital attenuator to the input impedance of the RF power transistor which was found to be $(7.78 - 11.26j)\Omega$ through a simulated load-pull (Figure 1).

The input matching circuit is shown in Figure 4.14. It is seen to consist of a matching network, bias-tee as well as a buffered amplifier to set the desired LDMOS gate voltage. The required reactance of **C1** and **L1** of Figure 4.14 can be calculated using Equations 39 and 40.

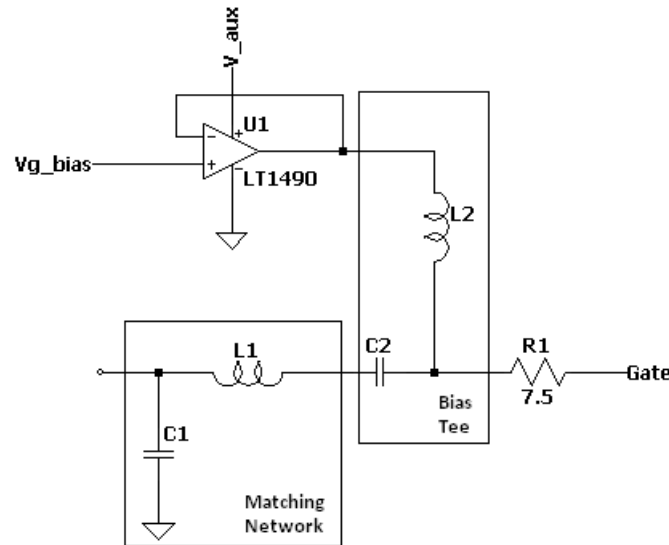


Figure 4.14: Schematic of the two element LC input matching network and bias-tee for the PA input section.

$$X_{C1} = -Z_0 \sqrt{\frac{R_L}{Z_0 - R_L}} \quad (39)$$

$$X_{L1} = \sqrt{R_L(Z_0 - R_L)} - X_L \quad (40)$$

Given $R_L = 7.78 \Omega$ and $X_L = -11.26j \Omega$, we obtain the following component values:

C1 = 43.75 pF and **L1** = 27.62 nH.

The values of **C2** and **L2** which form the bias-tee are the same as the values calculated for the bias-tee in the transceiver module.

Therefore **C2** = 500 pF and **L2** = 1 uH.

4.3.6 Output Matching Network

The final matching network to calculate for the PA module is the output matching network. Figure 4.15 shows the PA output stage matching network.

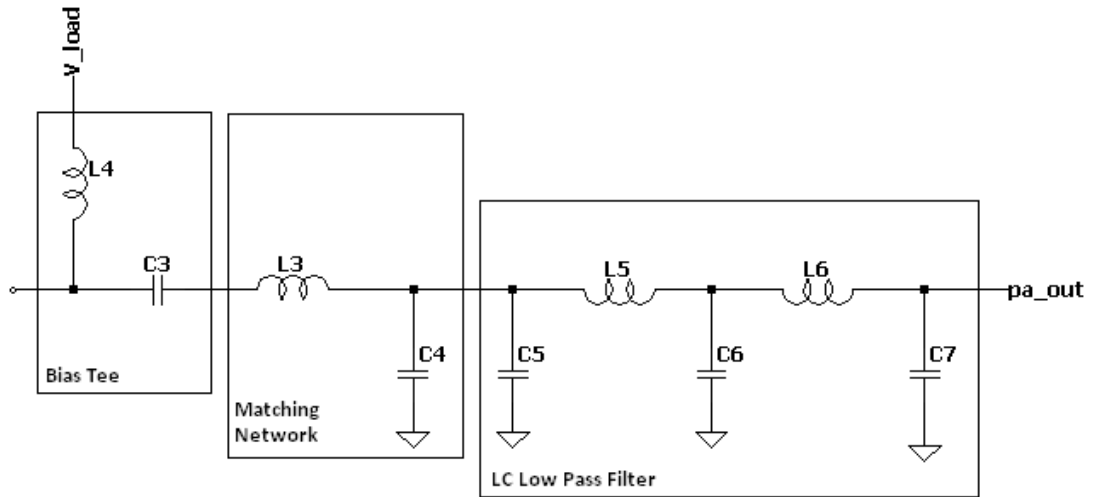


Figure 4.15: Schematic of the bias-tee, LC matching network and 5-th order LC low pass π -filter for the PA output section.

The required reactance of **L3** and **C4** of Figure 4.15 can be calculated using Equations 39 and 40.

Given:

$$R_L = 5.01 \Omega$$

$$X_L = -3.65j \Omega$$

we obtain the following:

$$\mathbf{L3} = 17.54 \text{ nH and } \mathbf{C4} = 56.28 \text{ pF.}$$

The bias-tee and fifth-order low pass π -filter retain the same values as calculated in the transceiver module section:

$L4 = 1 \text{ uH}$, $C3 = 500 \text{ pF}$

$C5$, $C7 = 30.5 \text{ pF}$, $C6 = 45.44 \text{ pF}$

$L5$, $L6 = 54.97 \text{ nH}$

However since $C4$ and $C5$ are in parallel (Figure 4.15), both capacitors can be replaced by a single capacitor of value 86.78 pF .

V_{load} is connected to V_{load} of Figure 4.1 where R_{load} represents the DC load of the PA output stage.

4.3.7 Harmonic Balance Simulation

With the synthesis of the required input and output matching networks and LC filters complete, it is now possible to simulate the complete power amplifier (PA) circuit to evaluate the power of the second- and third-order harmonics. The chosen simulation method is harmonic balance (HB) which was covered in the literature review section of the dissertation. The input side of the PA circuit is shown in Figure 4.16. A $+20 \text{ dBm}$ 50Ω single-tone power source is used to simulate the output of the Si4468 transceiver PA stage.

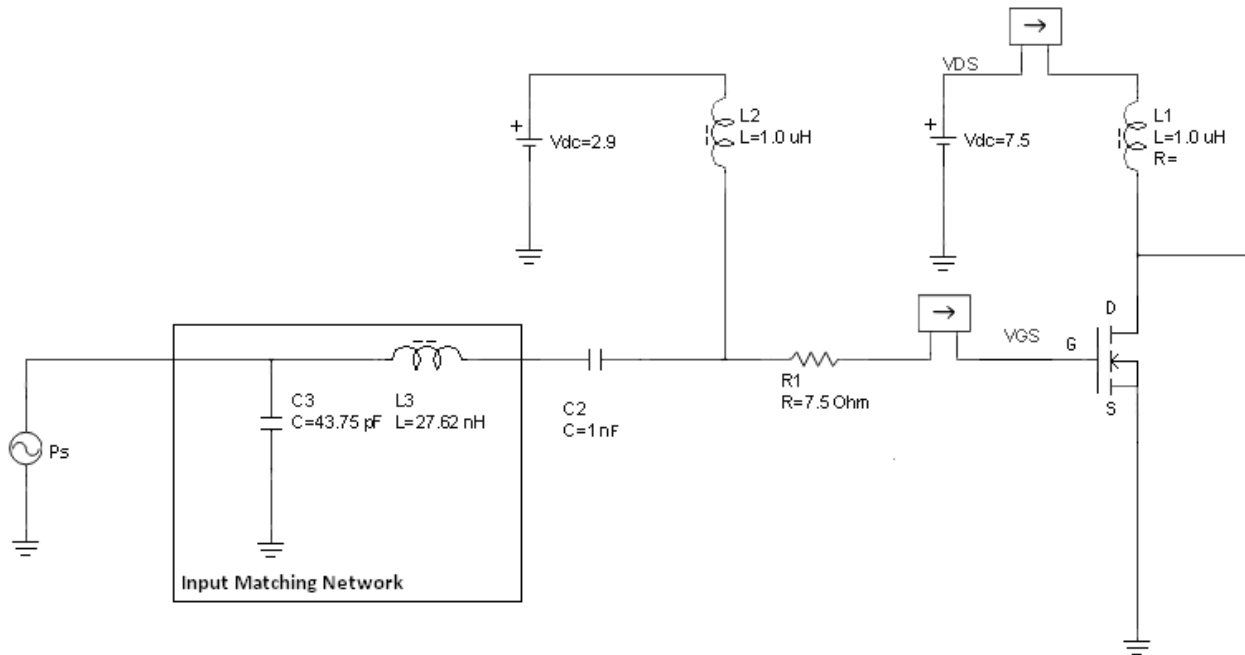


Figure 4.16: Simulated input network for the PA, including the LC matching network and bias-tee.

The resulting RF power amplifier output network is shown in Figure 4.17. However, the bias-tee inductor and load termination of 50Ω are not shown. The drain bias voltage for the HB simulation is set at 7.5V with a gate voltage of 2.9V (Figure 4.16). Current and voltage probes were added to measure the quiescent DC supply voltage as well as the voltage and current waveforms at the 50Ω load termination.

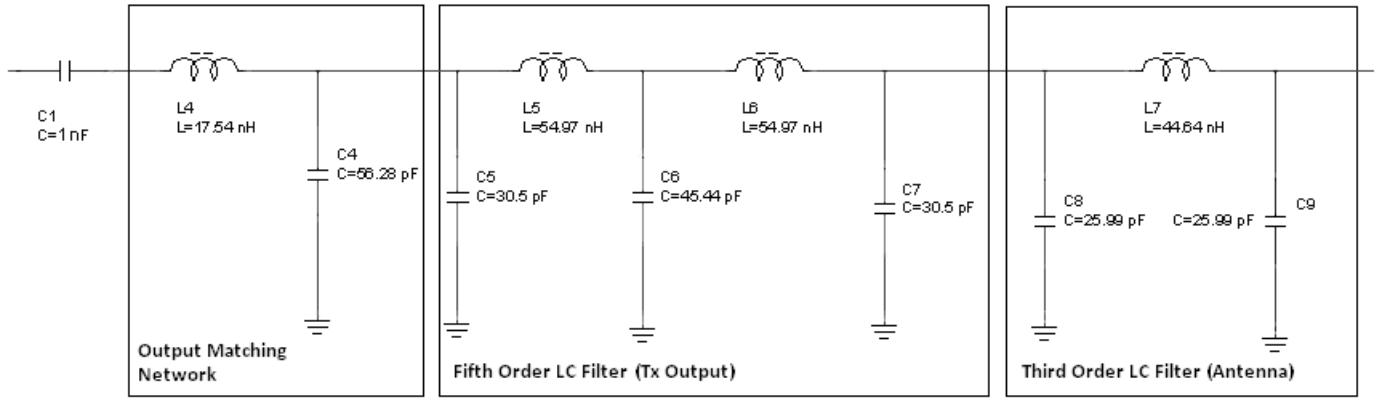


Figure 4.17: Simulated output network for the PA, including the LC matching network, fifth-order LC filter and additional third order, LC π -filter section.

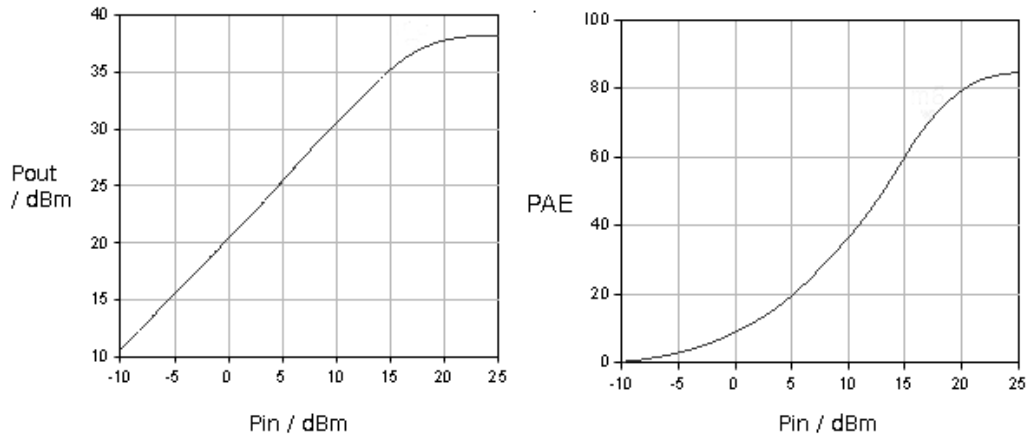


Figure 4.18: P_{out} , PAE vs P_{in} simulation for the complete RF power amplifier circuit using harmonic balance (HB).

Our first objective is to determine the RF input power to the PA at P1dB compression. Figure 4.18 are plots generated by a harmonic balance (HB) simulation of the complete PA circuit. It can be seen that at approximately +16 dBm of input power, the P1dB compression point of the PA is reached at a large signal gain of ≈ 19.5 dB (down from ≈ 20.5 dB in the linear region) at a PAE of around 70 %. It can be seen that decreasing the input power to the PA generally results in a sharp reduction in the PAE figure.

Figure 4.19 shows the results of the harmonic balance simulation at +16 dBm of CW input power. It can be seen that a peak power of +36.634 dBm at the fundamental tone has been achieved with a second-order harmonic tone at -43.917 dBm, which meets our 50 dBm attenuation design requirements.

These results indicate that the final stage, third-order LC filter is sufficient and that the filter order does not need to be modified. A quiescent DC supply current of approximately 900 mA was obtained, with a sinusoidal load current waveform.

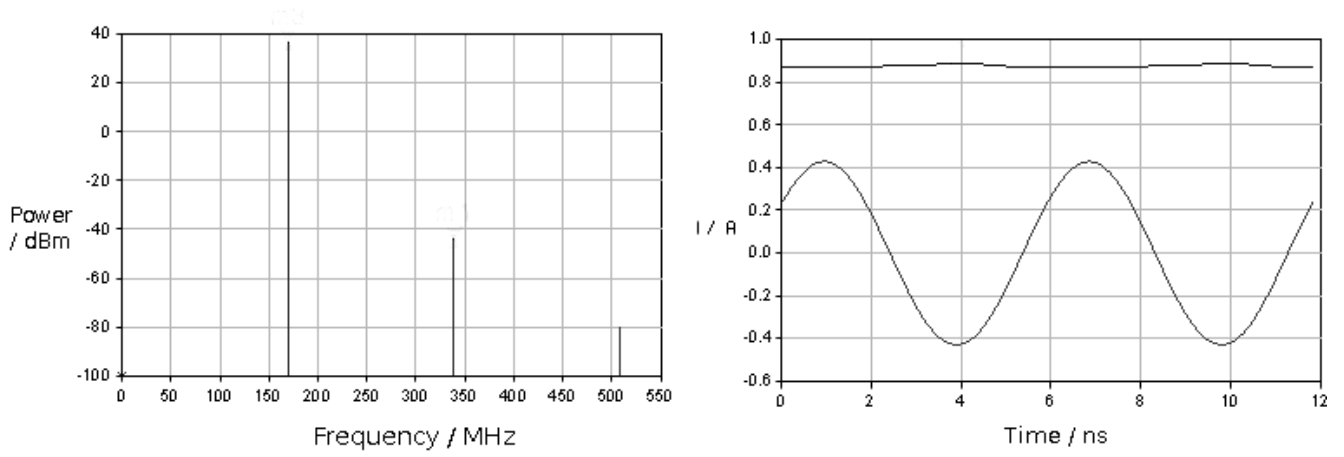


Figure 4.19: Simulation of the RF power amplifier output tones (shown on the left), with the drain current waveforms shown on the right.

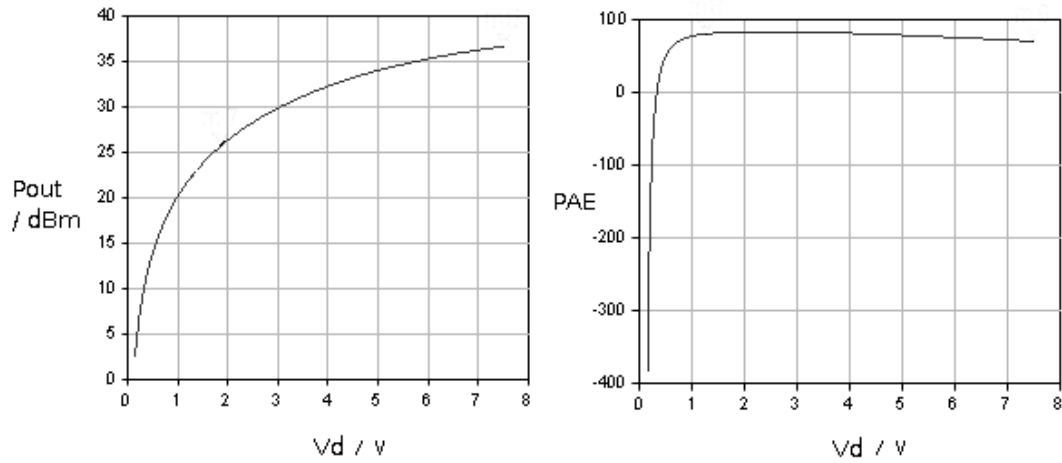


Figure 4.20: P_{out} , PAE vs V_D simulation for the complete RF power amplifier circuit using harmonic balance (HB).

Figure 4.20 shows the results of a harmonic balance simulation in which the output power and PAE were measured for different values of drain bias. It can be seen that the output power graph follows a logarithmic trend from $V_D = 0.15$ V to $V_D = 7.5$ V while the PAE figure tends to stay relatively constant with decreasing V_D until it decreases significantly at around $V_D = 0.5$ V.

Therefore figure 4.20 suggests that for a linear power ramp function, the step size per processing cycle would have to be smaller for a lower output power than for a higher output power set point in order to maintain accuracy using the drain power control strategy.

4.3.8 Drain Current Sense Circuit

As mentioned previously, a method to dynamically monitor the DC quiescent drain current of the RF power amplifier is required. A variety of current sense networks exist; however, a high-side current sense circuit with a current shunt and differential amplifier is widely used for this purpose as the circuit provides good results and is straightforward to design.

A typical high-side current sense circuit is shown in Figure 4.21. The differential amplifier used is the LT1490 rail-to-rail op-amp, which allows the output voltage to swing close to the ground rail.

Given $R_2 = R_3$ and $R_4 = R_5$, the transfer function of the circuit is shown in Equation 41:

$$V_O = I_1 \cdot R_1 \cdot \frac{R_5}{R_3} \quad (41)$$

The required shunt resistance (R_1) can be calculated given that the maximum voltage drop of R_1 ($V_{I_{max}}$) should be limited to 150 mV (to limit power dissipation) and that the maximum shunt current is never expected to exceed 1.5 A ($I_{I_{max}}$):

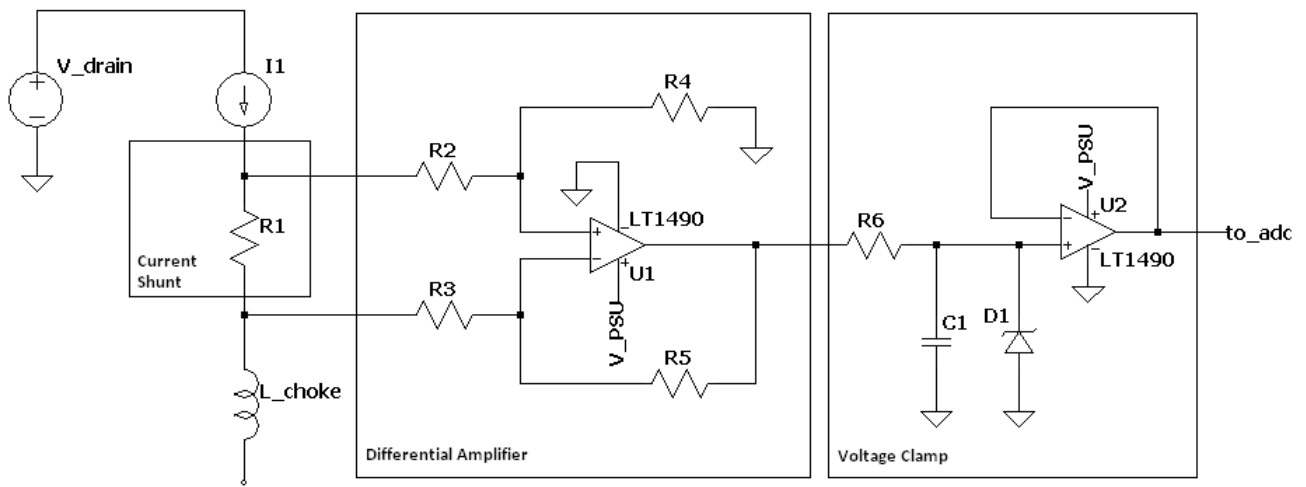


Figure 4.21: Current sense circuit used for monitoring the drain current of the RF power amplifier.

$$R_1 = \frac{V_{I_{max}}}{I_{I_{max}}} = \frac{150mV}{1.5A} = 100 \text{ m}\Omega \quad (42)$$

Next, the required gain of the differential amplifier can be calculated using equation 43, given that we would like to output 3.3V to our ADC when the current through R_1 is approximately 1.5A:

$$A_v = \frac{3.3V}{1.5A \cdot R_1} = 22 \quad (43)$$

We start by choosing a value for R_2 and R_3 :

$$R_2 = R_3 = 1 \text{ k}\Omega$$

Since $A_v = 30$, $R_4 = R_5 = 22 \text{ k}\Omega$

Note: In high side current sensing circuits that use op-amps, the supply voltage to the op-amp is required to be higher than the common mode voltage (CMV) [102] at the inputs.

Since R_4 is significantly higher than R_2 , $CMV = V_{\text{drain}}$ and thus the highest expected CMV is around 7.5V, which is lower than our PSU voltage. Next, it is possible to determine the component values of the voltage clamp circuit. **D1** functions as a voltage regulator so that the input at U2 never goes past the zener regulation voltage V_Z . The datasheet of the MCP3202 ADC states that the maximum input voltage to the ADC is equal to $V_{CC} + 0.6V$. Since $V_{CC} = 3.3V$, our input voltage should never exceed 3.9V.

Hence, we choose a $V_Z = 3.6V$ and a Zener diode with a 1 % tolerance. Assuming a zener regulation current $I_Z = 5 \text{ mA}$ (typical), the value of the zener current limiting resistor (**R6**) can be calculated given a maximum output voltage of 12V from the differential amplifier stage:

$$R_6 = \frac{12 - 3.6}{I_Z} \approx 1.6 \text{ k}\Omega \quad (44)$$

Finally the value of **C1** is not critical and is designed to filter out high frequency noise.

We therefore set **C1** = 1 nF.

4.3.9 Sensitivity Analysis

In order to determine how sensitive our PA design is to component tolerances a sensitivity analysis was performed. For each sensitivity test the values of all the passive components in the input and output network were adjusted by the specified percentage tolerance.

Table 2 shows the results of the sensitivity analysis. It can be seen that at a tolerance of $\pm 5 \%$ we get $\approx 1.44 \text{ dB}$ of variation in the fundamental tone (n=1) power and 7.51 % variation in the PA efficiency.

Tolerance / %	Tone n=1 / dBm	Tone n=2 / dBm	DC Current / A	Efficiency / %
-5	36.395	-40.552	0.798	73.62
+5	37.836	-44.526	1.232	66.11
-3	36.345	-42.013	0.819	72.56
+3	36.859	-44.921	1.001	65.01
-2	36.498	-42.706	0.836	71.855
+2	36.785	-44.708	0.955	67.061

Table 2: Results of the sensitivity analysis performed on the final PA design for different component tolerances.

As a general guideline, we would like less than 0.5 dB variation in the fundamental tone power and less than 10 % variation in the PA efficiency. These conditions are achieved for a component tolerance of $\pm 3 \%$. Overall, it is noted that the resulting PA design is not very sensitive to component tolerances. Thus, SMD components must be chosen with a tolerance smaller than or equal to 3 %. This restriction was applied when sourcing components to construct the physical modules that were designed.

4.4 Conclusion

This section covered the detailed design of three core subsystem modules, namely the power supply, transceiver, and power amplifier modules. For the power supply module, a linear voltage

regulation circuit was designed in order to provide a variable drain voltage from 0 to 7.5V, with a power dissipation capability of around 4W. This was followed by the transceiver module, for which the component values of the PA output and LNA input matching and filtering networks were calculated. Finally the power amplifier module was designed, with a simulated load-pull and stability analysis being performed, as well as the calculation of component values for the filtering and matching networks.

It was shown, through simulation, that a PAE of 76.29 % and large signal gain of 19.778 dB was achieved for the RF power amplifier at compression. For the stability analysis, a Rollet stability factor K of 1.31 was obtained at the design frequency, indicating unconditional stability of the RF power transistor. A stability analysis was also performed on the complete power amplifier circuit with a $K > 1$ obtained. Finally, a sensitivity analysis was performed in order to ensure that the design does not deviate significantly with component tolerance. It should be noted that although the performance of the system, through simulation, provides a theoretical guideline to measure against, it does not directly model non-ideal component characteristics which can lead to different results.

It therefore becomes necessary to develop a more advanced simulation model or simply iteratively construct a physical system prototype through measurements, which is the topic of the next section.

5 Design Implementation and Verification

This section will cover the physical implementation and design verification of the schematics that were developed in the high-level and detailed design and simulation sections. Therefore, the aim is to build a system prototype and determine whether the physical system meets the design requirements identified earlier. To this end, three separate categories of performance evaluation were created, including the power amplifier, output spectrum, and RF power control performance categories.

5.1 Laboratory Setup

Having completed both the high-level and detailed design phases, it becomes a requirement to perform design verification using a number of RF measurement instruments. To this end, a RF laboratory (shown in Figure 5.1) was used for research and development. The available instrument type and model for all instruments is indicated in Table 3.



Figure 5.1: RF laboratory used for research and development.

Type	Model
DSO (250 MHz)	Rigol DS1202Z-E
Vector Network Analyzer (1.5 GHz)	Siglent SVA1015X
Spectrum Analyzer (1.5 GHz)	Rigol DSA815-TG
Bench Multimeter	Rigol DM3068
Function Generator (3 MHz)	GW Instek GFG-8215A
Power Supply	Rigol DP832
Soldering Station	RS PRO 65W 202-4487
Hot Air Station	RS PRO 700W 124-4133
Workbench Lamp	Duratool 15W LED

Table 3: Table of laboratory test equipment used for testing.

The most commonly used test equipment during development was the Rigol DSA815 spectrum analyzer, Rigol DM3068 multimeter and the Rigol DP832 power supply. Furthermore, the

coaxial cable type that was utilized for interconnects was RG-174/U mainly due to its flexibility and acceptable path loss characteristics. For all types of measurement, the relevant laboratory test equipment used was warmed up for approximately 30 minutes before measurement to thermally stabilize each instrument. This was required to produce repeatable measurements, additionally Figure 5.2 shows the soldering materials that were available in the lab.

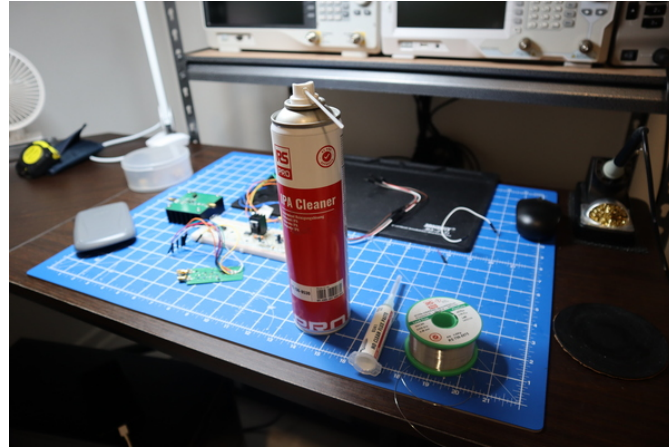


Figure 5.2: IPA solvent, MG-8341 flux and Pb-free Solder.

5.2 System Prototype

The main RF blocks that were physically constructed are shown in Figure 5.3. It is seen to consist of separate RF and analogue/digital sections (designed and built from scratch), which is required to maintain signal integrity. The analog/digital section implements all of the non-RF circuitry discussed in this dissertation and, therefore, can be placed on a breadboard. The RF circuitry makes use of gold plated female-to-female SMA connectors to chain the RF modules together. Standard jumper wires are then used to connect the RF blocks to the analogue/digital section on the breadboard (Figure 5.4).

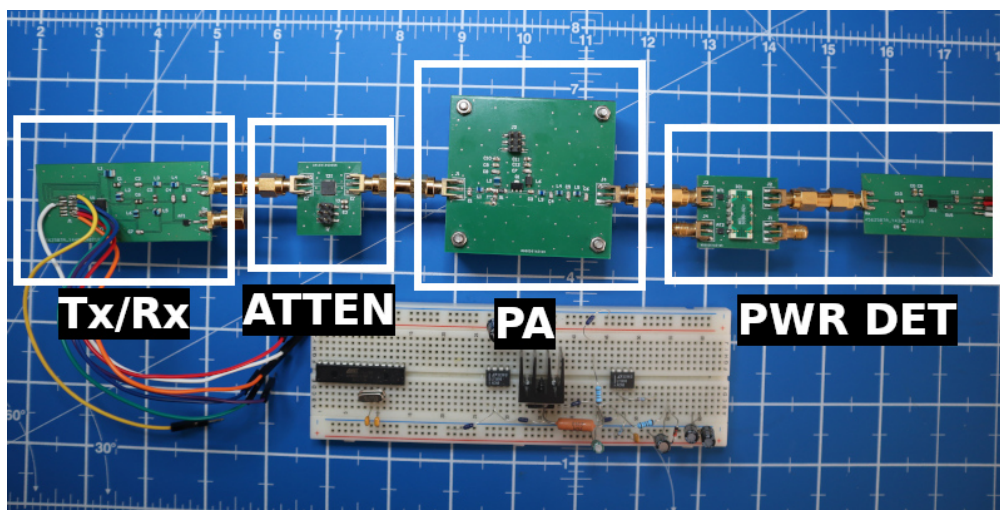


Figure 5.3: System prototype showing the: transceiver, attenuation, RF power amplifier and RF power detection PCBs which are chained together using SMA connectors (Figure 3.1).

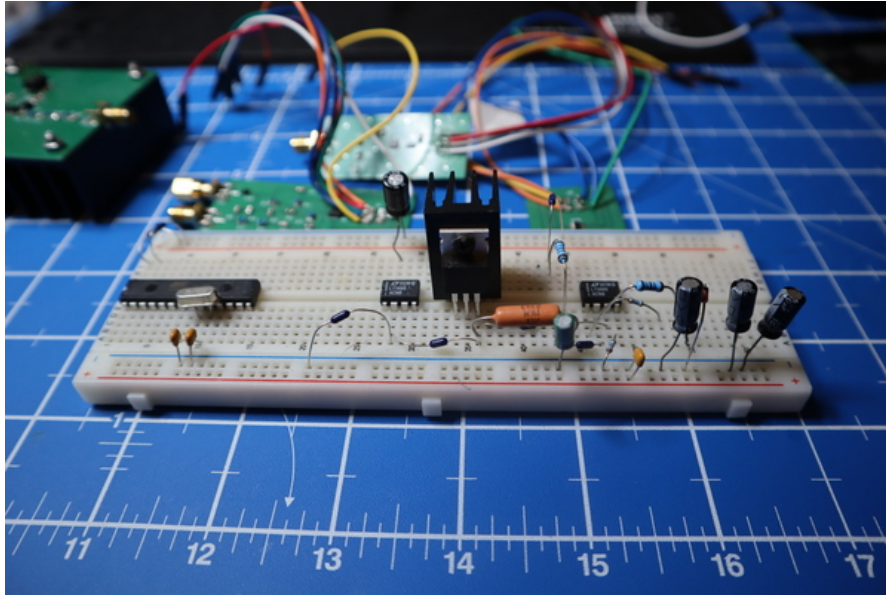


Figure 5.4: Analogue and digital section containing the power supply, current sense and drain voltage control circuitry.

The main AC power MOSFET for the PSM (Power Supply Module) is shown in the middle of Figure 5.4. This MOSFET corresponds to M1 in Figure 4.1 of the drain bias control circuit, which is responsible for the linear regulation of the drain output voltage. As mentioned in the high-level design section, most of the wasted power in M1 is dissipated as heat. Therefore, a heatsink is required to ensure that the operating temperature is within an acceptable limit. The electrolytic capacitors shown in Figure 5.4 are the 10 μF capacitors that are used to smooth the DAC / ADC outputs and inputs to obtain less noisy measurement samples. Finally, the large orange resistor is the 100 m Ω current shunt shown in Figure 4.21 with the microcontroller shown on the left with a 16 MHz quartz crystal.

5.3 Power Amplifier Module

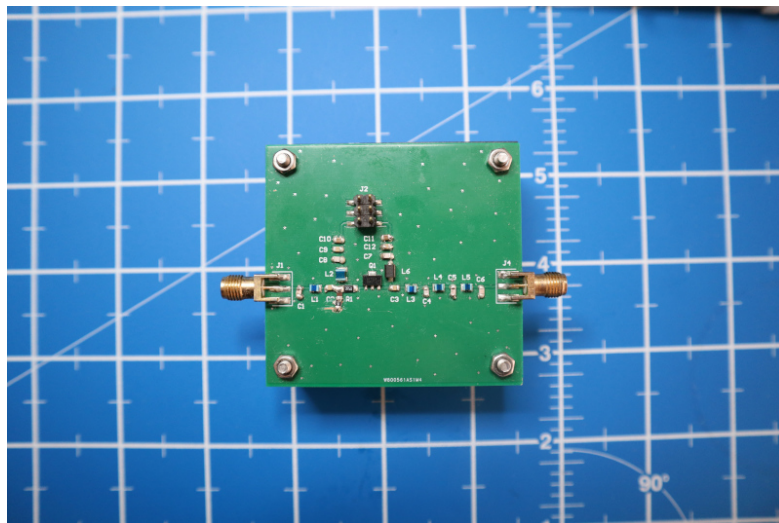


Figure 5.5: Resulting physical RF power amplifier module that was built, with SMA female connectors used at the input and output port of the RF module.

The resultant RF power amplifier module that was constructed is shown in Figure 5.5. Murata branded 0805 SMD RF inductors and capacitors were mainly chosen during component selection due to the availability of an online tool, supplied by Murata, called 'Simsurfing' [103] that contains an extensive database of component characteristic data (including 4 port S-parameter). Simsurfing was used to evaluate the frequency performance of each LC component before placing a formal order to ensure that the SRF, Q-factor and current handling capability of each component was sufficient to implement the practical design. It is worth mentioning that these three parameters are critical and cannot be ignored by a designer when selecting a real component.

Once the components and PCB's for the PA module were delivered, the components were then hand soldered (using a 65W temperature controlled soldering iron and 700W hot air station) onto the corresponding PCB using lead-free solder and MG Chemicals 8341, no-clean flux. Despite the name 'no-clean', these types of resin/rosin hybrid flux do leave a sticky residue on the PCB after soldering. The advantage of a no-clean flux is that the residues are noncorrosive, which ultimately slows down/prevents PCB corrosion. It is usually a good idea to clean any flux residues with isopropyl alcohol or IPA after soldering. High-quality precision tweezers were used to place the components onto the PCB prior to soldering.

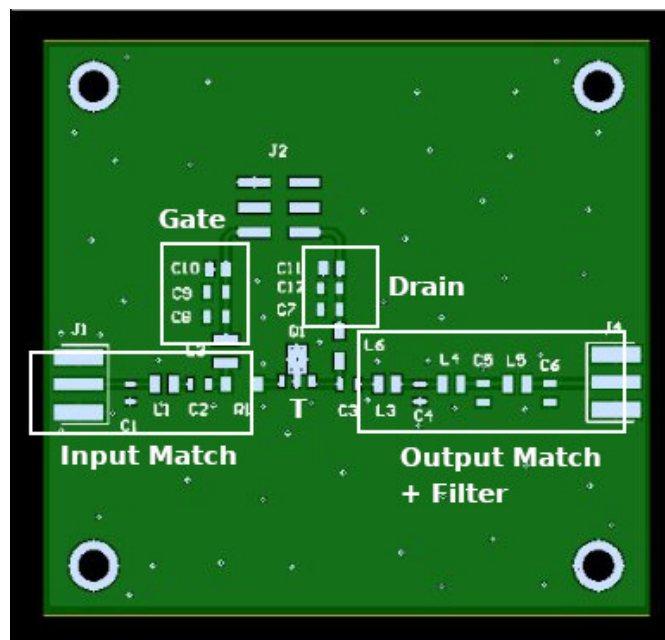


Figure 5.6: Detailed view of the PCB that was designed for the power amplifier module, with all the relevant subcircuits annotated.

The different sections of the power amplifier PCB are shown in more detail in Figure 5.6. It is seen to consist of an input matching section, output matching, and filtering section, followed by the gate and drain voltage bias sections. Four M3 sized mounting holes are used to attach the heatsink to the PCB while via stitching is required to provide a low impedance to ground across the entire plane of the PCB in order to reduce the effects of ground loops and provide a stable AC and RF ground reference. The PCB layout of the power amplifier corresponds to the schematics of Figures 4.14 and 4.15. It should be mentioned that the symbol T denotes the placement of the RF transistor in Figure 5.6. The transistor source pad also includes via stitching to sink heat

down into the ground plane. It is common practice to use large PCB ground planes as heat sinks.

5.3.1 Requirements

The main design requirements for the power amplifier module are as follows:

- 169.4 MHz (VHF) operating frequency.
- Stable output ($K > 1$).
- +30 dBm of RF RMS output power achieved.
- PAE greater than 60 %.
- Large signal gain (G_{LS}) of at least +10 dB.

Through design simulation, the following parameters were obtained (Table 4):

min(PAE) (3 % tolerance) / %	G_{LS} / dB
65.01	19.5

Table 4: Theoretical PAE and G_{LS} (Table 2).

These results are, of course, theoretical and may differ from the performance figures obtained from the practical implementation of the design. For example, the PAE figure from Table 4 was obtained by modeling discrete inductors and capacitors with infinite or theoretical Q which results in no resistive losses. In practice, RF inductors and capacitors have a finite Q and therefore the results in Table 4 do not reflect the performance of the real-world system. Essentially, the system prototype is expected to have a reduced efficiency and a large signal gain, primarily due to parasitic losses in the system as mentioned.

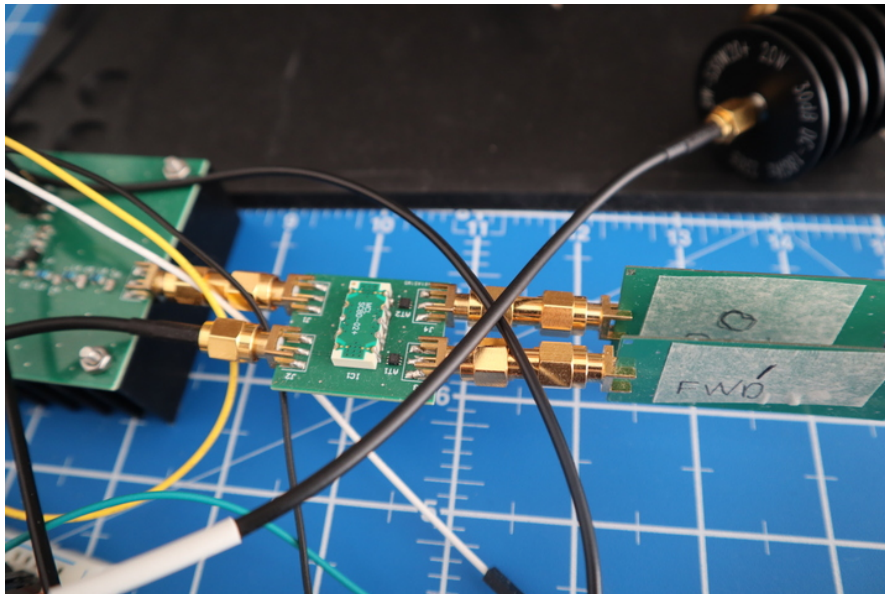


Figure 5.7: Directional coupler and RMS-log power detector section for the RF forward power, with the PA output being fed into a 30 dB passive attenuator.

Laboratory measurements are therefore required to validate these set of design requirements against the final practical design. Figure 5.7 depicts the coupling path at the output of the

RF power amplifier which is used to determine the RF forward power. The directional coupler consists of 4 distinct ports which are terminated with a female SMA connectors. The ports on the left hand side of the RF directional coupling PCB are the RF in and out references, while the ports on the right are the forward and reverse reflected references. In the case of Figure 5.7 both forward and reverse RMS power detectors are connected, allowing us to determine VSWR and when a non $50\ \Omega$ load is attached. However for VARPA, only the forward RMS power detector is utilized for output power control.

5.3.2 Test Procedure

Two laboratory procedures are described to determine the real-world PAE and G_{LS} of the overall system. The laboratory procedure for determining PAE will now be described. To determine PAE we will use Equation 9. This requires us to know the RF input power to the PA (P_{in}), which is defined as the output power of the transceiver module (Tx). The theoretical output power of the transceiver module is +20 dBm but there are other parasitic losses which have not been accounted for, since we are assuming ideal operating conditions.

Step 1 of the procedure is to measure the RF output power of the transceiver module using a Rigol DSA815 spectrum analyzer. The Rigol DSA815 (Figure 5.8) is an entry-level spectrum analyzer with a bandwidth of 1.5 GHz and a maximum RF input power of +20 dBm. Therefore, it is not possible to directly feed the output of the transceiver module into the DSA815, since there is a risk of overloading the RF front-end and potentially damaging the instrument. Therefore, a 20 W +30 dB passive attenuator (Figure 5.9) will be placed in line to reduce the RF input power to the Rigol DSA815 for all subsequent power measurements.

Once P_{in} is determined a measurement of P_{out} is required at 30 dBm of RF output power. The measurement of P_{out} is very similar to P_{in} , which also makes use of the +30 dB attenuator, with the difference being the output port that is measured.

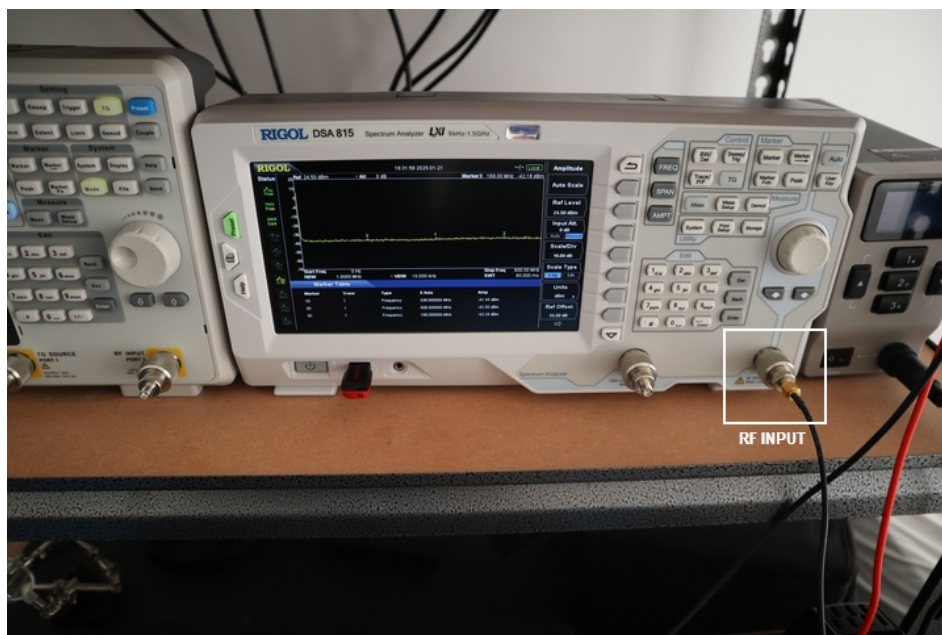


Figure 5.8: Rigol DSA815 spectrum analyzer that was used for all power spectrum measurements, with the RF input port indicated in the figure.

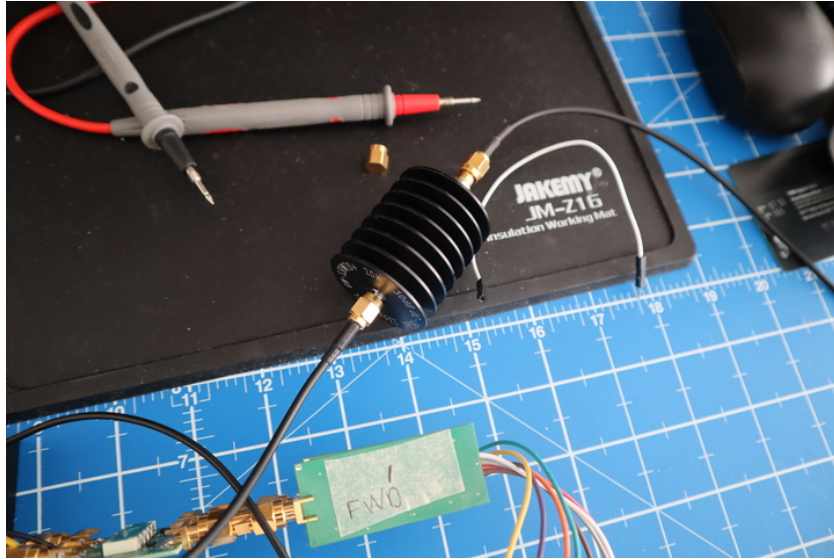


Figure 5.9: 30 dB passive inline attenuator used to reduce the signal power before being fed into the spectrum analyzer.

Finally, once P_{in} and P_{out} are determined, the final step in the calculation of the PAE for the system involves determining the total DC power consumed by the power supply. The power supply used for testing is the Rigol DP832, which is a 3 channel 32 V / 3A DC supply. Most Rigol instruments have ethernet or networking capability, which allows us to retrieve screenshots for reporting purposes. The Rigol DP832 has the capability of showing the total power drawn per channel in watts (W). It should be noted that the total DC power consumed is a combination of the PA DC input power and the power drawn/dissipated by the power supply module, which includes linear regulators. We will therefore have to isolate the power dissipation of the power supply module from the RF power amplifier itself.

At this point, the real-world PAE can then be determined, as well as G_{LS} by using the measured input and isolated PA output power.

5.3.3 Results

Figure 5.10 is a screenshot that depicts the output spectrum of the system, configured through the serial interface, at +30 dBm. Screenshots are dumped from memory on the DSA815 via the internal SCPI interface over ethernet.

A reference level offset of 33 dB was chosen to account for the passive attenuator of 30 dB (real attenuation measured at 31.3 dB) and 1.7 dB of measured path loss in the RG174/U coax and connectors that were used. It can be seen that an output power (P_{out}) of +30.11 dBm was achieved. An extra 0.11 dB of power gain was achieved, which is slightly above the 30 dBm requirement. This indicates that we have achieved sufficient power gain in relation to the transceiver output and that the performance of our matching networks is adequate. The RF detector mode used for power measurements on the Rigol DSA815 was set to 'max-peak' which is typically used for CW power measurements.

Next, the output power of the transceiver module (Tx) must be determined, which represents P_{in} to the power amplifier module (PA). Table 5 indicates the 3 separate RF output power readings that were obtained on the DSA815 for the transceiver output. These 3 readings are averaged to

yield a P_{in} of 19.22 dBm. This output power figure indicates that our PA matching network for the transceiver module is also performing optimally and that we are more or less, achieving the expected RF output power from the transceiver.

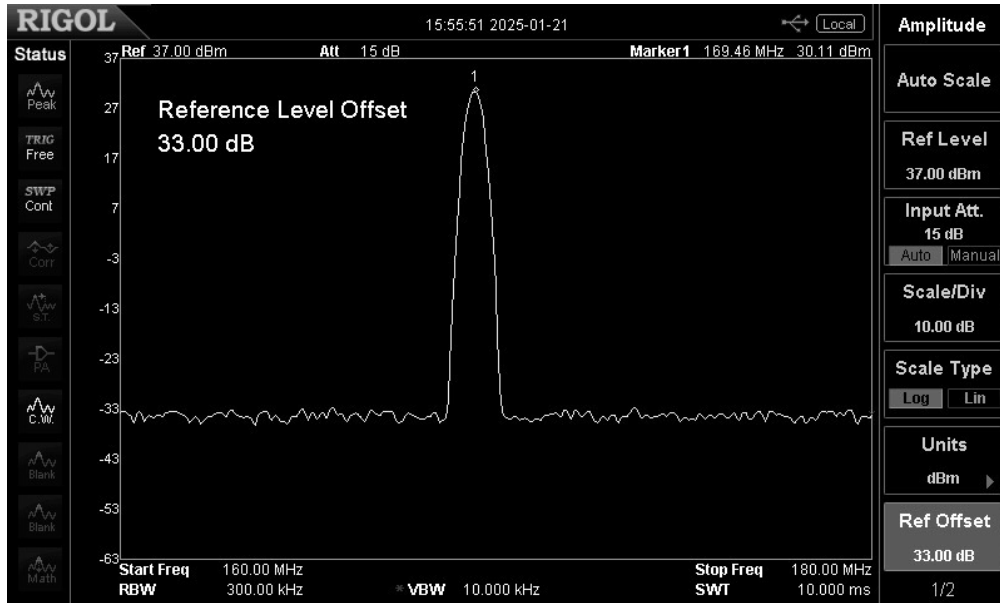


Figure 5.10: PA output spectrum @ 3000 cdBm set point.

TX Output (dBm)
19.20
19.24
19.21

Table 5: Transceiver (Tx/Rx) output power for all three readings which were finally averaged.

Furthermore, the output spectrum of the transceiver was measured, with a maximum nth-order harmonic at around -24.41 dBm of power. This means that we have achieved more than 40 dB of attenuation relative to the carrier after the filtering stage in the transceiver module. This indicates that our filtering network is performing adequately.

The gain (G_{LS}) can now be calculated using the difference between P_{out} and P_{in} , resulting in a large signal gain of around 10.89 dB. Therefore, our large signal gain requirement has been met, in this case, as defined in the design requirements section above. It is worth mentioning that this is not the gain capability of the PA module itself, as the power gain of the PA is dependent on the drain supply voltage.

Maximum gain is typically achieved at maximum supply voltage. However, our PA module intentionally limits the drain voltage using the RF power detection feedback loop in order to achieve a specific output power set point (30 dBm). More specifically, the drain voltage at the RF MOSFET and RMS RF power detector output are sampled using a pair of ADCs and then compared to the setpoint. The DAC controlling the drain voltage is then adjusted in discrete steps until our power setpoint is reached.

Finally, the DC input power was obtained for the power amplifier module at a +30 dBm setpoint using the lab procedure described and is shown in Table 6.

Power Supply Module / W	P_{DC} Power Amplifier / W
0.75	1.82

Table 6: P_{DC} for the power supply module (PSM) and power amplifier (PA).

P_{DC} therefore is equal to 1.82W. Converting P_{in} and P_{out} to watts and substituting in Equation 9 produces a PAE figure of around 54 % at +30 dBm of RF output power. This is in contrast to the theoretical efficiency of 65.01 % obtained in Table 2 for 3% component tolerance (worst case). As mentioned previously, some decrease in efficiency is expected due to resistive losses in the components used and the non-ideal Q. Additionally, it was shown in Figure 4.20 that the PAE decreases for decreasing drain voltages. Since we are operating at +30 dBm of output, we are decreasing the drain voltage significantly with around a $V_d = 3V$ required to achieve 1W of RF output power.

However, this is still lower than our required PAE of 60%. Therefore, modifications to the design would have to take place in order to improve efficiency, most likely by increasing the Q of the matching networks. However, for VARPA achieving an efficiency greater than 60 % was not the primary design objective of the project.

Regarding PA stability, a series of low-frequency spurs were observed intermittently at around 7.5 MHz, at around -17 dBm of RF power upon power up, indicating poor stability. After an investigation, the source of the problem seemed to have originated from the incorrect value of gate damping resistor (4.5 Ω used instead of 7.5 Ω). Once the correct value of damping resistor was used, no low or high frequency spurs were observed on the output power spectrum (Figure 5.11) which indicates sufficient PA stability ($K > 1$).

5.4 Output Spectrum

5.4.1 Requirements

The primary requirement of the PA output spectrum was to achieve 50 dBc of attenuation relative to the carrier at +30 dBm of RF output power in order to suppress harmonics and spurs. This ensures a clean sinusoidal output tone with minimal external interference, which is important for various emission testing requirements. However, the process of commercialization is not to be underestimated, as a variety of additional legal/technical regulations and compliance issues would have to be addressed in the design, which is beyond the scope VARPA. The design of VARPA attempts to produce reasonable performance at a low cost given the initial design requirements laid out at the beginning of the dissertation, which by themselves are guided by existing RF short range device regulations.

5.4.2 Test Procedure

The experimental procedure is rather straightforward and involves connecting the filtered output of the RF power amplifier to the DSA815 with the 30 dB attenuator in line and measuring the output spectrum.

5.4.3 Results

The output spectrum of the PA module at +30 dBm of output power is shown in Figure 5.11. The primary tone (169.4 MHz) and the nth-order harmonics are clearly visible. Three distinct markers are used to annotate the second and third-order harmonic tones, as well as the carrier. Using marker 2D which has an amplitude of -24.84 dBm, we have achieved 54.92 dB of attenuation relative to the carrier (marker 3D).

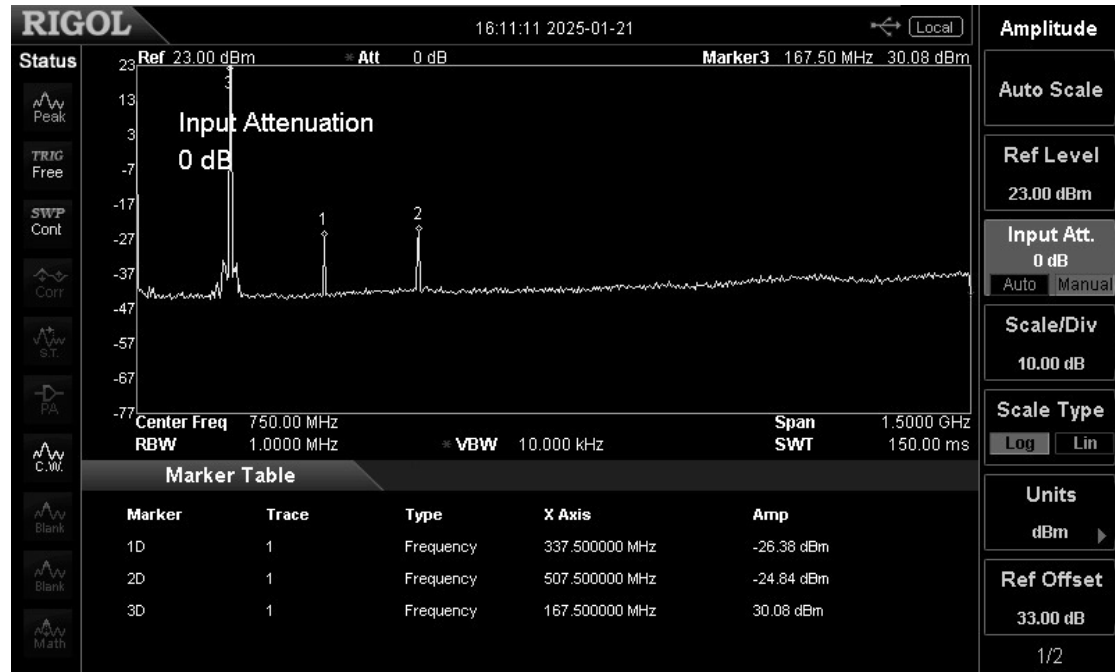


Figure 5.11: Wideband output spectrum of PA at a +30 dBm set point (for fifth-order low-pass output π -filter).

Generally, it is the second- and third-order harmonics that are of primary concern since the amplitude of these tones tends to decrease rapidly with increasing order. Therefore, only the second and third-order harmonic tones are displayed in Figure 5.11. It can be seen that the third order harmonic (marker 2D) has a slightly larger output power than marker 1D which represents the second order harmonic, which indicates that the PA design is slightly balanced. No higher order harmonics are present according to the output spectrum. However, the 50 dB harmonic attenuation target has been achieved, which was an important requirement of the system.

Given these results, the design requirements for this section have been validated. It should be noted that the exact frequencies of the markers as shown in Figure 5.11 are not precise across the full frequency span of the analyzer due to a limited number of sampling points used by the instrument.

The average temperature of the PA heatsink was also measured using an inexpensive digital IR thermometer. The PA module was set to output 30 dBm as performed earlier. The PA was given time to reach thermal equilibrium before being measured. After 2 hours of CW output, a temperature of 34.56 degrees Celsius was recorded, indicating that the PA design is also thermally stable.

5.5 RF Power Control

In this dissertation, two power control strategies were explored. The first method involved varying the drain bias voltage of the LDMOS power transistor, which is fed by the linear voltage regulation circuit that was designed. The second second involved varying the attenuation of the 30 dB F1956NBGI8 digital attenuator that was placed at the input of the PA module.

5.5.1 Requirements

The requirements for the RF power control methods explored in this dissertation were as follows:

- 0 to 30 dBm RMS power output range.
- 0.5 dB power step size, within 20% error.
- < 1.5 dB absolute amplitude accuracy of RF output power.
- Power step ramp settling time < 1s.

It is worth mentioning that it is generally easier to achieve a consistent delta in the RF output power rather than reaching an specific value of RF output power each time. Absolute amplitude accuracy typically requires a calibration source or procedure, which must be applied periodically to retain absolute amplitude accuracy. The amplitude accuracy error arises because of thermal drift and other non-ideal, time-varying parasitic effects. Since the design does not include such a source, only changes in RF power are recorded and compared to the requirements. The 1.5 dB absolute amplitude accuracy is a rough figure that just ensures that the absolute power measured is not too far from the set point in relative terms (DSA815 amplitude accuracy).

5.5.2 Test Procedure

The laboratory testing procedure simply involves connecting the PA output to the DSA815 spectrum analyzer and then interacting with the system through the serial interface that was developed as part of the system firmware. This serial interface allows the desired output power to be set in step sizes of 0.5 dB. The output power level and power spectrum can then be analyzed with the DSA815 and recorded for further analysis.

5.5.3 Results

Figure 5.12 shows the serial output when the system is powered on/initialized or reset. The serial terminal operates at 19200 baud to improve data transfer rates. The number of bytes of available memory (SRAM) is displayed to keep track of stack, data, and heap usage within the firmware. A list of commands is displayed with a total of 8 commands available to the user.

A user of the system first issues a "PROBE" command which gradually ramps the gate voltage until the desired quiescent drain current is reached. In our case, this is approximately 100 mA. A 'probe' must be performed when starting a power ramp, since different drain voltages used for power control lead to different quiescent currents given a fixed gate voltage and therefore, the gate voltage must be determined dynamically. Figure 5.13 is the serial output of the "PROBE" command which indicates the desired drain current Q-point, probe settling time, and final drain quiescent current.

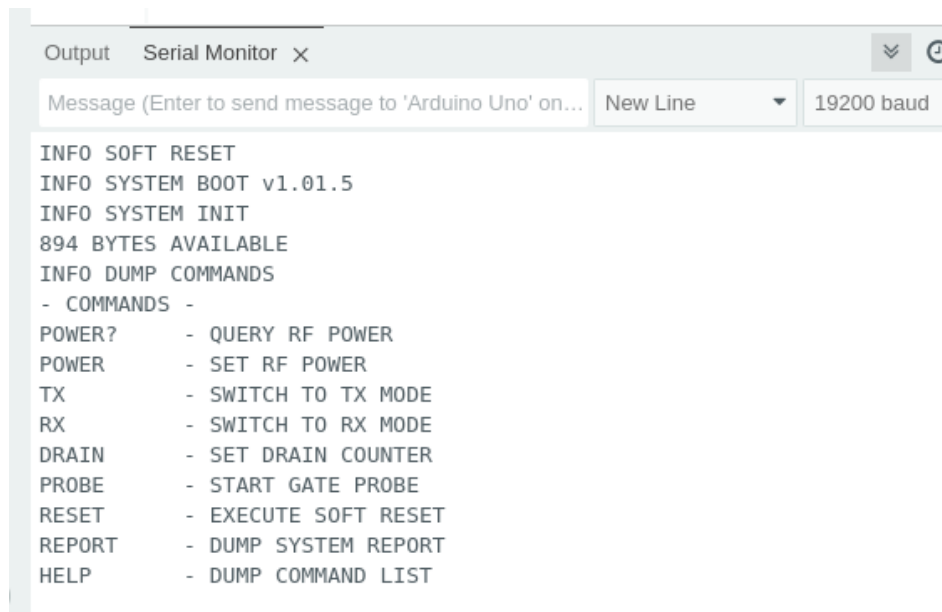


Figure 5.12: Serial command interface after boot up and the execution of the "HELP" command.

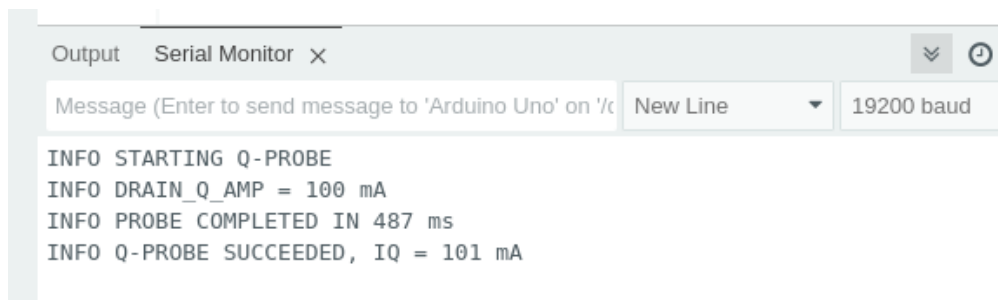


Figure 5.13: Serial output following a successful gate probe procedure (issued via the "PROBE" command) which is required before setting the output power.

A command line application (Figure 5.14) was developed that allows a user to graphically visualize the operation of the system rather than relying on commands issued through a serial terminal. The program allows a user to save and restore state information as JSON files as well as select the power control strategy (drain or attenuation based) used by the system. It is seen to consist of three main panes.

The gate and drain panes are used to report on the active biasing conditions of the RF power transistor, while the system pane is used to retrieve the overall status of the system, as well as the serial link status. Some functionality shown in the image (such as VSWR reporting) has yet to be implemented and is left as a task for future work.

If the probe was successful, a user may proceed by issuing a "POWER" command followed by the target power set point in cdBm as well as the power control strategy, all separated with a space. Here, cdBm represents centi-dBm values. Therefore, a value of 1000 cdBm represents a value of 10.00 dBm. The power set-point must be a multiple of 50 (0.5 dBm), which is a limitation built into the firmware in order to achieve the desired 0.5 dB target step resolution in output power.

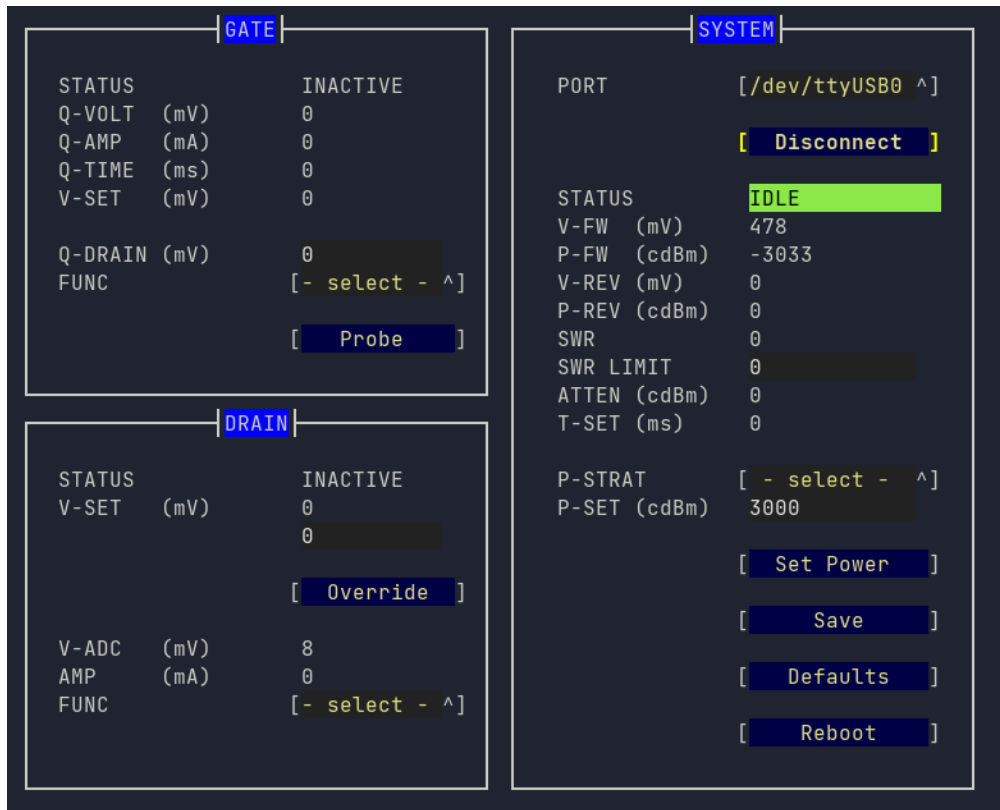


Figure 5.14: Interface for the terminal application used to interface with the system prototype. Only the essential features of the application have been developed.

The power control strategy can be either 0 for drain voltage control or 1 for attenuation-based control. Figure 5.15 shows the output following the issuing of 3 separate power commands. It should be noted that the power control algorithm used is simply a linear sweep function rather than a complex PID-based strategy, since the sweep settling time is well below the required threshold ($< 1s$).

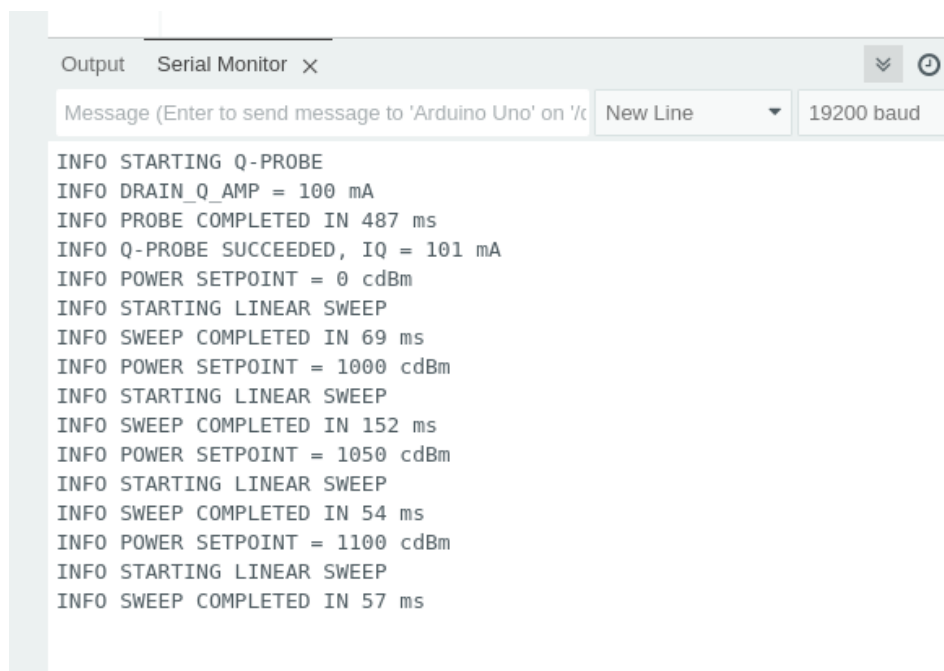


Figure 5.15: Serial output following a series of "POWER" system commands.

The output spectrum when the output set point is set to 0 cdBm is shown in Figure 5.16. Most of the time, a very accurate set point power relative to DSA815 was achieved, with a maximum error of 3.2% for 50 set points that were tested. For output power levels below 10 dBm the output power set point was found to be within 4% of the set point and within 6% for output power levels above 10 dBm. This is most likely due to the increased number of RMS power detector calibration points for power levels below 10 dBm.

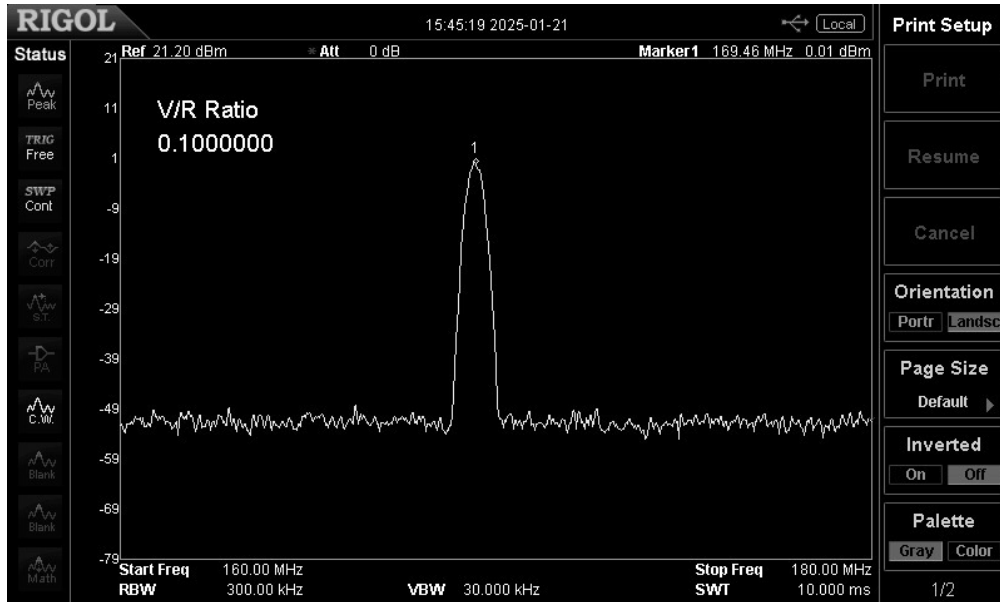


Figure 5.16: Output spectrum with 0 cdBm set point.

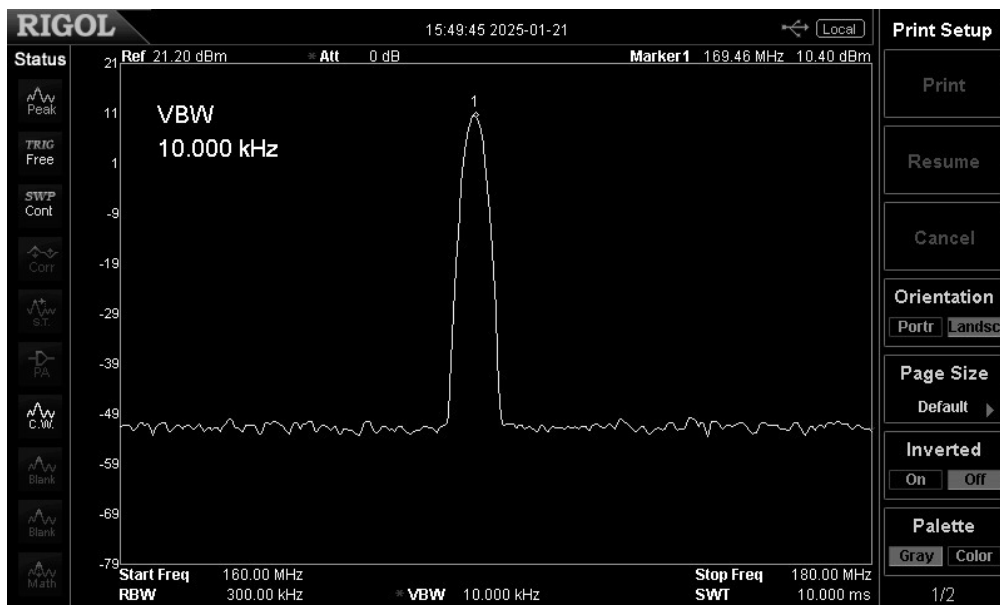


Figure 5.17: Output spectrum with 1000 cdBm set point.

In general, RF power set points that are a small distance away from the current set point will settle much quicker than set points that are farther away. The result of the "POWER" command will be the desired set point in cdBm as well as the total sweep time to reach the set point. Figure 5.17 depicts the power spectrum following the command "POWER", with a set point of 1000 cdBm (10.00 dBm). The absolute amplitude accuracy is around 0.4 dB in this case, which is well below

our 1.5 dB requirement. This procedure was repeated 50 times for different set points, resulting in a maximum amplitude accuracy of 0.9 dB, which meets our design requirement.

Figure 5.18 depicts the output spectrum when the "POWER" command was issued with the set point increased to 1050 cdBm (10.50 dBm). The most important factor in this case is the change in power compared to Figure 5.17, since we have increased the set point by 0.5 dB. As can be seen, the power delta is equal to 0.52 dB which is less than 5 % different from our set point of 0.5 dB. This was repeated for around 50 different increments (of varying sizes in cdBm) with the largest normalized deviation in step size of around 0.55 dB, which represents a 10% error. In all cases, the power ramp settling time was less than the required 1 s.

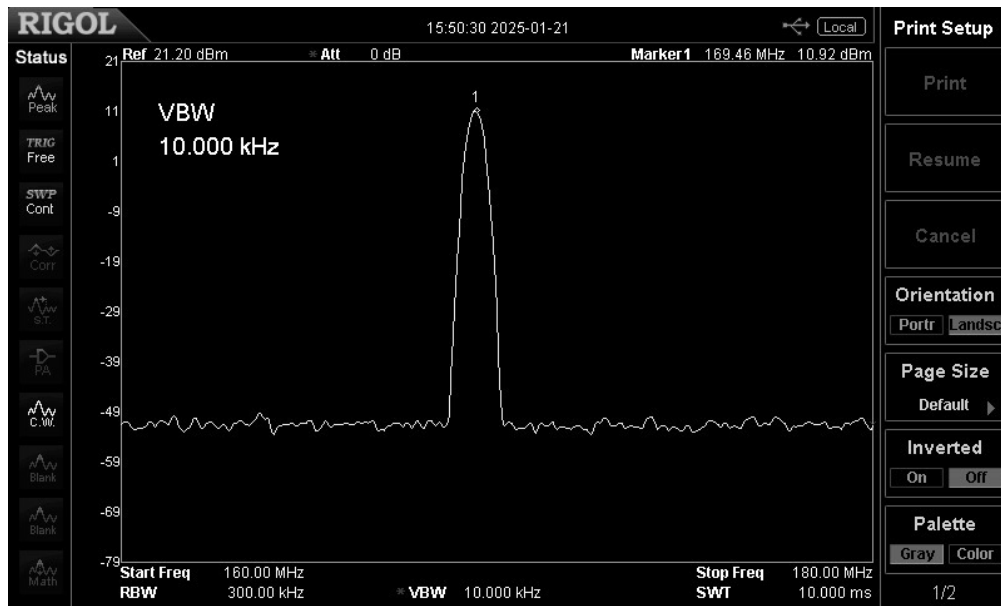


Figure 5.18: Output spectrum with 1050 cdBm set point.

The measured output power vs set point for both power control strategies is shown in Figures 5.19 and 5.20. Figure 5.19 in particular are for the first power control strategy (variable drain voltage), while Figure 5.20 is for the second power control strategy (variable input attenuation).

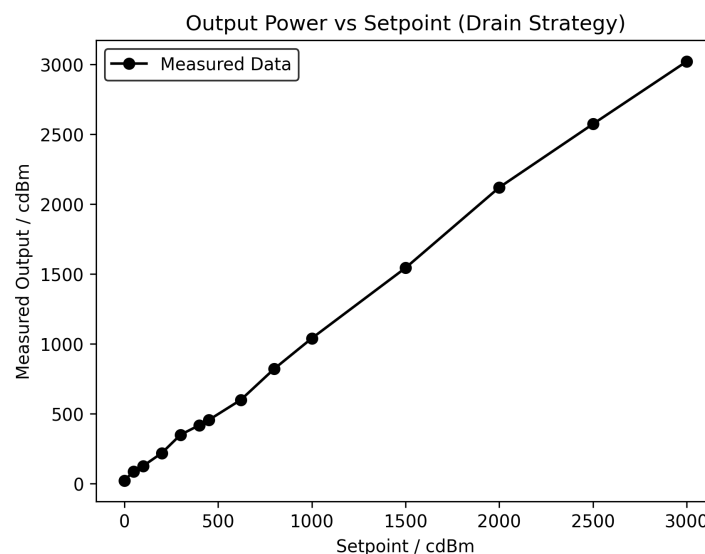


Figure 5.19: RF output power vs set point for the drain power control strategy.

As can be seen in Figure 5.19 the measured output power is roughly proportional to the target set point, but the trend is clearly not perfectly linear. However, this is to be expected since earlier simulations demonstrated that the relationship between P_{out} and drain voltage is not a simple linear function (Figure 4.20), and that the rate of change of output power, for lower power levels, is much higher and therefore more difficult to control.

The trend depicted in Figure 5.20 is clearly more linear and this is to be expected since the actual transfer function of P_{out} to input attenuation is roughly linear.

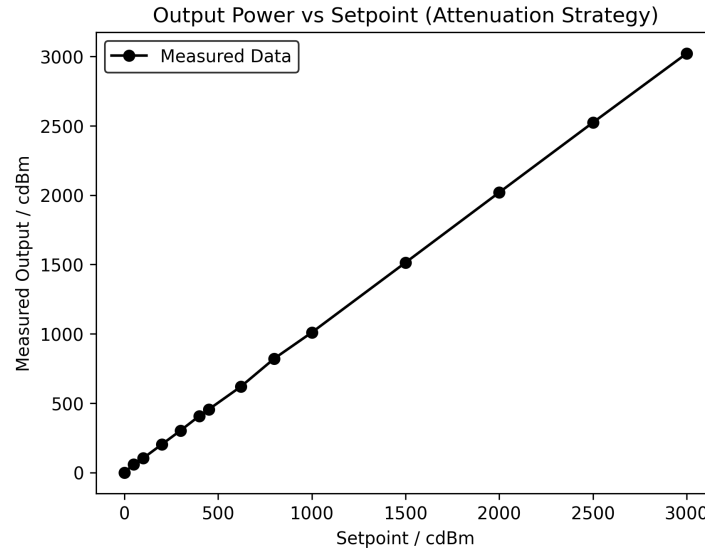


Figure 5.20: RF output power vs set point for the attenuation based power control strategy.

It should be mentioned that a system report may be generated using the "REPORT" command, shown as part of the help menu in Figure 5.12. An example of the generated report is shown in Figure 5.21. As shown, a detailed status report of the systems operating conditions is generated. This report is sent to the command line application periodically and in serialized form to update the terminal display.

According to the data collected, all four requirements have been met for each power control strategy, with a power step size error margin less than or equal to 20% of 0.5 dB which is considered acceptable, a maximum amplitude accuracy of 0.9 dB and a maximum settling time (T_S) of 550 ms in the worst case for both power control strategies.

Table 7 is a summary of the results obtained for both power control methods through measurement for 50 dBc step sizes and 1000 dBc step sizes. Furthermore, the maximum step size deviation is indicated by ΔD .

Strategy	Max T_S (50 cdB) / ms	Max T_S (1000 cdB) / ms	Max ΔD / cdB
Drain Control	20	450	8
Digital Attenuator	37	110	3

Table 7: Max Settling Time T_S and Step Size Deviation ΔD .

It can be seen that the drain power control method produced the lowest T_S for a step size of 50 dBc, while the attenuation control produces the lowest T_S for a step size of 1000 dBc and

also produced the lowest step size deviation ΔD . In both cases, the step size deviation is within the 20%, ΔD requirement. Table 8 is a summary of the total cost of components for each power control strategy. It can be seen that the drain control strategy is significantly cheaper to implement than the power control method using the digital attenuator.

Strategy	Total Cost (2024) / \$
Drain Control	2.51
Digital Attenuator	7.52

Table 8: Total component cost matrix in USD for each power control strategy.

```

Serial Monitor x
Message (Enter to send message to 'Arduino Uno' on '/c' New Line 19200 baud

21:51:13.321 -> INFO DUMP SYSTEM REPORT
21:51:13.353 -> - GATE -
21:51:13.353 -> STATUS : PROBE_OK
21:51:13.353 -> Q-VOLT (mV) : 2800
21:51:13.384 -> Q-AMP (mA) : 1320
21:51:13.384 -> Q-TIME (ms) : 23
21:51:13.416 -> V-SET (mV) : 2800
21:51:13.416 -> Q-DRAIN (mV) : 1000
21:51:13.448 -> - DRAIN -
21:51:13.448 -> STATUS : INACTIVE
21:51:13.448 -> V-ADC (mV) : 0
21:51:13.480 -> V-SET (mV) : 0
21:51:13.480 -> AMP (mA) : 1320
21:51:13.480 -> - SYSTEM -
21:51:13.512 -> STATUS : IDLE
21:51:13.512 -> V-FWD (mV) : 3300
21:51:13.512 -> P-FWD (cdBm) : 5680
21:51:13.544 -> P-SET (cdBm) : 0
21:51:13.544 -> T-DELTA (ms) : 0

```

Figure 5.21: Serial output following the "REPORT" command which dumps a system status report.

5.6 2FSK Tx and Rx

Since both the transmit and receive circuits for the Si4468 transceiver were developed in the detailed design section, we can perform a series of simple tests to verify that 2FSK data packets can be sent and received successfully between two transceivers under different operating conditions (range, output power). For the tests, the system prototype shown in Figure 5.3 was utilized as a transmitter while a second, standalone transceiver module (Tx/Rx) and controller was built and utilized in receive (Rx) mode.

A radio configuration header file, with the modulation-specific requirements defined in Section 3.1, was generated using WDS3 (Wireless Development Suite 3) by Silicon Labs and then flashed onto both system controllers. For 2FSK testing it was necessary to replace the $50\ \Omega$ dummy load with a real antenna to carry out range testing at different power levels. To this end, a simple half-wave dipole for each transceiver was constructed out of 18 AWG electrical wire and then tuned/trimmed to resonance at 169.4 MHz using the Siglent SVA1015X vector network analyzer (VNA). The resulting input impedance ($43\ \Omega$) of the antenna is not a perfect $50\ \Omega$ match but yielded a VSWR < 2 which was sufficient for testing.

Since the Rx transceiver requires less than 50 mA of supply current to operate, a 9V battery can be used as a power source, which then allows the receiver (Rx) to be separated from the transmitter

at different ranges for testing. The VARPA firmware for the Tx transceiver was then modified to continuously transmit a 2FSK data packet (instead of a CW tone) with a 'magic' header, while the Rx firmware was modified to emit an audible tone (using a small piezo buzzer) every time a data packet was successfully received and decoded.

For the actual experiment itself, the transceiver was placed at four different distances from the transmitter, ranging from 100 m to 2 km for four different Tx power output levels. It should be mentioned that the experiment was performed in a dense urban environment with multipath effects, with the net effect of reducing the link range. Table 9 shows the results of the experiment, with an effective range of around 1.2 km observed.

Range / km	0 dBm	10 dBm	20 dBm	30 dBm
0.1	Audible	Audible	Audible	Audible
0.5	Silent	Silent	Audible	Audible
1	Silent	Silent	Silent	Audible
2	Silent	Silent	Silent	Silent

Table 9: Effective range testing for different output power levels.

The experiment was then repeated, but with a 2FSK data rate calculator integrated into firmware. Table 10 indicates the results of the rate tests. It can be seen that the theoretical rate of 33 Kbps is never actually achieved, most likely because data packets are dropped/lost as a result of multipath effects. Nonetheless, a data rate greater than 25 Kbps is maintained for distances less than 1 km, which represents a 25 % reduction in data rate compared to the theoretical rate.

Range / km	0 dBm	10 dBm	20 dBm	30 dBm
0.1	30.2 Kbps	32.5 Kbps	32.3 Kbps	31.7 Kbps
0.5	0 Kbps	0 Kbps	26.5 Kbps	28.9 Kbps
1	0 Kbps	0 Kbps	0 Kbps	25.1 Kbps
2	0 Kbps	0 Kbps	0 Kbps	0 Kbps

Table 10: Effective data-rate testing for different values of output power and range.

5.7 Conclusion

This section has evaluated three main system performance benchmarks, which were: RF power amplifier performance and the associated PA output spectrum as well as the performance of the RF power control circuitry, against the research and design objectives described in the introduction of this dissertation. For the PA module category, it was shown that the PA module had good overall performance characteristics, despite achieving a PAE of 54% compared to the 60% design target, and the presence of stability issues that were later fully rectified.

The output spectrum analyzed was shown to provide adequate harmonic suppression as per design requirements, but not enough to meet EU ERC standards for short range devices. Thus, the additional 3 element LC π -filter shown in Figure 4.7 should be included in the final design. However, utilizing additional filtering stages runs the risk of increasing the insertion loss at the output of the PA, but the extent of this would have to be determined empirically through experimentation.

Finally, two RF power control strategies were evaluated. It was shown that the attenuation-based power control strategy produced the most linear set point transfer function with the smallest settling time for large power steps, while the drain voltage-based power control method produced a less linear, but acceptable transfer function with the advantage of a significantly reduced total component cost.

6 Main Conclusion

This dissertation involved the research and design of a software adjustable RF power amplifier as well as an investigation into effective RF power control strategies that can be used for LDMOS power transistors operating in the VHF band. It was successfully demonstrated through simulation and experimentation that all research and design objectives, with the exception of the 60 % PAE efficiency target, outlined at the beginning of the dissertation, were met. The system was also designed and built from the ground up, using no external hardware modules or software libraries.

The only outstanding question is which power control strategy is most 'effective' from both a cost and performance perspective. The total component cost required for the drain power control strategy is significantly lower than most commercially available 0.25 dB digital step attenuators, requiring only a MOSFET, op-amp and a few passives (Table 8). However, this is at the cost of reduced linearity of the output power vs the desired set-point. Therefore, it is up to the end user to decide whether cost, linearity, or settling time are more important, since this will ultimately decide which power control strategy to choose for their particular application.

According to the performance data collected for the power amplifier module, all four power control performance requirements were met for both power control strategies that were tested, with a maximum power step size deviation ΔD of less than 20 %, a maximum amplitude accuracy of 0.9 dB and a maximum settling time (T_S) of 550 ms in the worst case for both power control strategies. In summary, the system was able to output a 169.4 MHz CW tone with an output power ranging from 0 to 30 dBm in 0.5 dB steps, with a PAE of ≈ 54 % and a large signal gain of 10.89 dB. These were the main achievements of the project.

It is worth mentioning that most modern PA designs can benefit from using RF transistors that are internally matched, eliminating the need for external matching networks and load-pull simulations. However, these wide-band power transistors generally suffer from lower efficiencies compared to the narrow-band performance that unmatched transistors provide, since the PAE generally increases with a high matching network Q-factor which is only achievable with an external matching network.

6.1 Design Limitations

Almost all the given design requirements could be considered limitations as an end user always appreciates improved system performance. However, a notable limitation is the limit in output power of around 30 dBm. This is more a firmware limitation than a hardware limitation. Theoretically, the P1dB compression point could be reached by applying the right bias to the LDMOS transistor, however most regions across the world do not typically operate above 1W in non-commercial settings and most regions limit RF output power to 500 mW for short-range devices.

Another notable limitation in the system is the 0.5 dB step size limit in output power. This limitation primarily stems from the limit in ADC resolution used to sample the output power from the RMS logarithmic power detector. In order to decrease this 0.5 dB limit, a higher resolution ADC would have to be used, and possibly a different power control algorithm involving some type of PID controller. However, 0.5 dB is typically acceptable in the vast majority of cases.

It is also worth mentioning that the power dissipated in the linear regulator (power supply module) which is responsible for setting the PA supply/drain voltage is significant and a redesign using a variable switching regulator could be explored to ensure that switching noise has no significant effect on 2FSK data transmission and reception.

6.2 Future Work

Future work might involve improving the performance of the PA design in terms of efficiency and accuracy, as discussed above, as well as adding new features to the firmware and software utilities. Examples of software features include adding VSWR reporting to the CLI application, as well as the ability to switch off the RF output if no active load is connected for an extended period of time. Modifying the PA output filtering section is also another area of interest if the product is to meet EU and South African radio emission requirements.

END OF DISSERTATION

7 References

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8 Appendix A

All source code and PCB project files that were created for VARPA can be found at <https://github.com/lunar-rf/varpa>

```
% Passthrough Frequency %
Fp = 169.4
% Ripple (dB) %
Ap = 0.5
Z0 = 50
start_freq = 169;
end_freq = 200;
freq_inc = 0.5;
% Optimization Variables %
opt_x = 100
opt_freq = 0
final_filter = 0

num_steps = (end_freq - start_freq) / freq_inc + 1;

for i = 1:num_steps
    fp = start_freq + (i - 1) * freq_inc;
    filter = rffilter("FilterType","Chebyshev", ...
        "ResponseType","Lowpass", ...
        "FilterOrder", 5, ...
        "Implementation","LC Pi", ...
        "PassbandFrequency",fp*10^6, ...
        "PassbandAttenuation",Ap)

    zin = gamma2z(gammain( filter . sparameters (Fp*10^6)))
    if abs(Z0 - zin) < opt_x
        opt_freq = fp
        opt_x = abs(Z0 - zin)
        final_filter = filter
    end
end

frequencies = linspace(0,2*Fp*10^6,1001);
rfplot(final_filter , frequencies)
```

Listing 1: MATLAB LC low pass π -filter design script.